

Module:4 Measurement of Industrial Parameters

Measurement of temperature: thermistor and LM35, Measurement of pressure: strain gauge and piezoelectric type, Measurement of distance: ultrasonic, linear variable differential transformer and capacitance type, proximity sensor, Infrared sensor, Pulse oximeter and Tachometer.

Measurement of temperature

Thermistor:

Thermistor (THERMally sensitive resISTOR) are non-metallic resistors (semiconductor material), made by sintering mixtures of metallic oxides such as manganese, nickel, cobalt, copper and uranium.

Thermistors have a Negative Temperature Coefficient (NTC), i.e. resistance decreases as temperature rises. Thermistors follow the principle of decrease in resistance with increasing temperature. The material used in thermistor is generally a semiconductor material such as a sintered metal oxide (mixtures of metal oxides, chromium, cobalt, iron, manganese and nickel) or doped polycrystalline ceramic containing barium titanate (BaTiO_3) and other compounds. As the temperature of semiconductor material increases the number of electrons able to move about increases which results in more current in the material and reduced resistance. Figure shows a graph of resistance vs temperature for a thermistor.

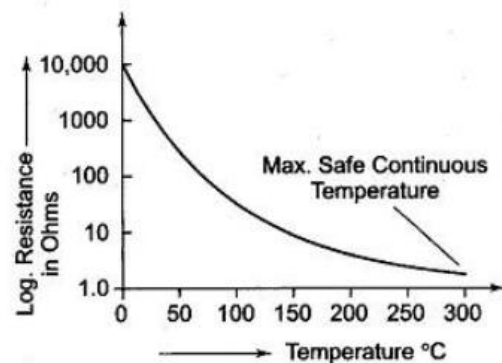


Fig. Resistance vs Temperature Graph of a Thermistor

The mathematical expression for the relationship between resistance of a thermistor and temperature is

$$R_{T_1} = R_{T_2} \exp \left[\beta \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \right]$$

R_{T_1} = resistance of thermistor at temp. T_1

R_{T_2} = resistance of thermistor at temp. T_2

β = constant depending upon the material of thermistor. it changes in the range of 3500-4500

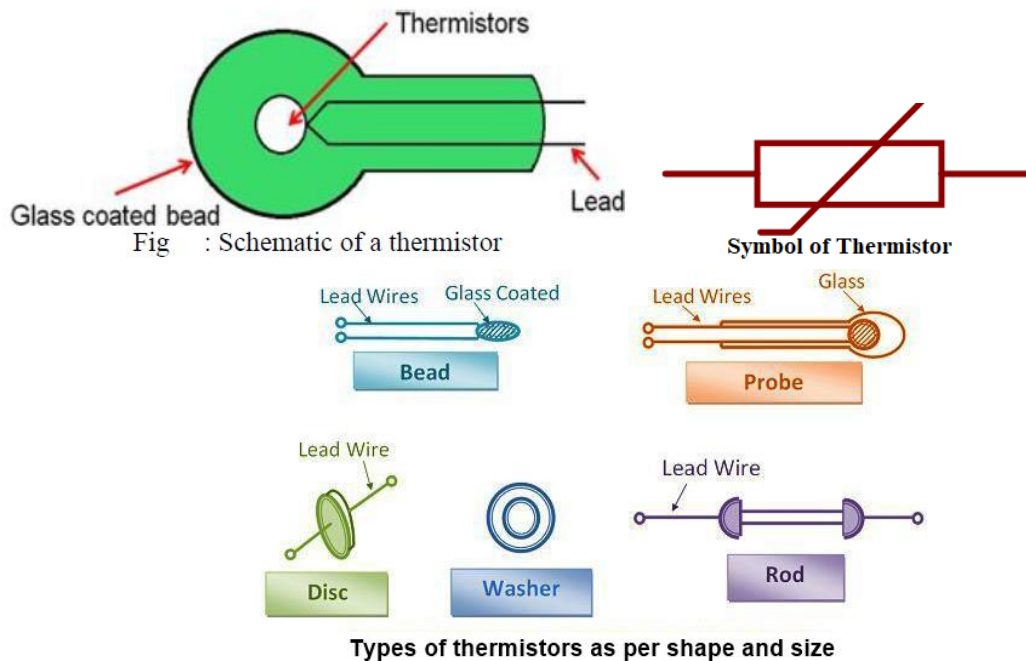
T_1 is initial temp and T_2 is the final temp.
and changes in resistances is R_{T_1} and R_{T_2} at these temp.

This is the Non-linear relationship between resistance and temp

exp= exponential

The resistance at room temperature (25°C) for typical commercial units ranges from 100 Ω to 10 Ω . Thermistors cannot be used to measure high temperatures compared to RTDs. The temperature range at this working is -55°C to +150°C. In some cases, the resistance of thermistors at room temperature may decrease by 5% for each 1°C rise in temperature. This high sensitivity to temperature changes makes the thermistor extremely useful for precision temperature measurements, control and compensation.

Thermistors are rugged and small in dimensions. They exhibit nonlinear response characteristics. Thermistors are available in the form of a bead (pressed disc), probe or chip. Figure 41 shows the construction of a bead type thermistor. It has a small bead of dimension from 0.5 mm to 5 mm coated with ceramic or glass material. The bead is connected to an electric circuit through two leads. To protect from the environment, the leads are contained in a stainless steel tube. Typical thermistor configurations are as shown in Fig.



Properties of Thermistor:

- ✓ They have negative thermal coefficient. i.e. resistance of the thermistor decreases with increase in temperature.
- ✓ They are made up of the semiconductor materials.
- ✓ They are more sensitive than RTD and Thermocouples.
- ✓ Their resistance lies between 0.5Ω to $0.75\text{ M}\Omega$.
- ✓ They are generally used in applications where measurement range of temperature -60°C to 15°C .

Advantages:

- ✓ High sensitivity
- ✓ Can be used at normal room temperature
- ✓ High sensitivity
- ✓ Small size
- ✓ Low cost
- ✓ Fast response
- ✓ Simple conditioning circuit

Dis-advantages:

- ✓ Highly Non-linear in nature
- ✓ High sensitivity allows the thermistor to work at low temperature range
- ✓ Not suitable for wide temperature change
- ✓ Shielded cable have to be used

Applications:

- ✓ Applications include temperature measurements, compensation and control
- ✓ Used in air conditioners
- ✓ Used In detection of fire alarms

Example 25.16. A thermistor has a resistance temperature co-efficient of -5% over a temperature range of 25°C to 50°C . If the resistance of the thermistor is $100\ \Omega$ at 25°C , what is the resistance at 35°C .

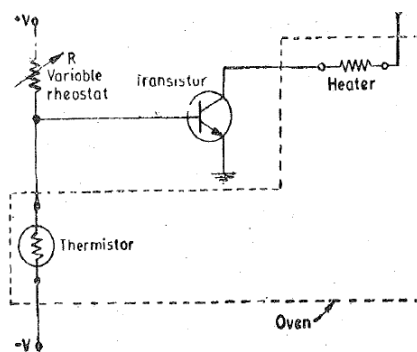
Solution. Resistance at a temperature of 35°C is :

$$R_{35} = 100 [1 - 0.05(35 - 25)] = 50\ \Omega$$

Example 25.18. An automatic temperature control arrangement for an electric oven is shown in Fig. 25.25. The thermistor has a resistance of $2000\ \Omega$ at 70°C . A potentiometer is used for obtaining balance conditions. The temperature of oven changes and therefore the rheostat has to be set at 1800°C to obtain balanced conditions. Using the linear approximation for the resistance temperature curve obtain the change in the value of resistance. The resistance temperature co-efficient can be assumed as $-0.05\ \Omega/\Omega-^\circ\text{C}$.

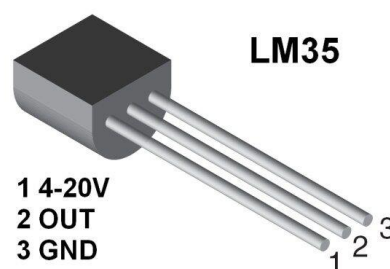
Solution. Using linear approximation (Eqn. 25.33), we have :

$$1800 = 2000[1 - 0.05(\theta - 70)] \text{ or } \theta = 72^\circ\text{C}.$$



LM35:

The LM35 is a precision integrated-circuit temperature sensor fabricated using semiconductor technology. It is an analog, linear temperature sensor whose output voltage varies linearly with change in temperature. It consists of a temperature-sensitive transistor arrangement built on a silicon chip, along with signal conditioning and amplification circuitry. The device is packaged in standard IC packages such as TO-92, SOIC, or TO-220. It has three terminals: **V_{CC} (supply voltage)**, **V_{OUT} (output voltage)**, and **GND**. The sensor is factory-calibrated in degrees Celsius, eliminating the need for external calibration. It can measure temperature from -55 degree celsius to $+150$ degree celsius. The voltage output of the LM35 increases 10mV per degree Celsius rise in temperature. LM35 can be operated from a 5V supply and the stand by current is less than $60\ \mu\text{A}$. The pin out of LM35 is shown in the figure below.



Working Principle of LM35:

The LM35 operates on the principle that the base-emitter voltage of a silicon transistor varies linearly with temperature. Inside the IC, the change in temperature alters the transistor characteristics, which are processed by the internal circuitry to produce a proportional output voltage. The LM35 provides a linear output of 10 mV per °C, meaning the output voltage increases by 10 millivolts for every 1°C rise in temperature. For example, at 25°C the output is approximately 250 mV. This linear and directly calibrated output makes the LM35 easy to interface with microcontrollers and analog circuits for temperature measurement.

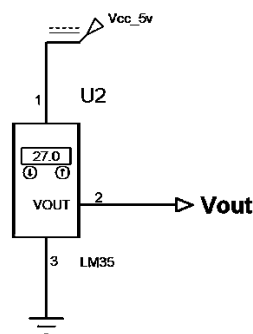
The output voltage increases by **10 mV per °C**:

- 0°C → 0 V
- 25°C → 250 mV
- 100°C → 1.0 V

Because of its linear output and direct calibration in Celsius, the LM35 can be easily interfaced with analog circuits and microcontrollers for accurate temperature measurement.

LM35 Interfacing Circuit

As such no extra components required to interface LM35 to ADC as the output of LM35 is linear with 10mv/degree scale. It can be directly interfaced to any 10 or 12 bit ADC. But if you are using an 8-bit ADC like **ADC0808** or **ADC0804** an amplifier section will be needed if you require to measure 1°C change. LM35 can also be directly connected to Arduino. The output of LM35 temperature can also be given to comparator circuit and can be used for over temperature indication or by using a simple relay can be used as a temperature controller.



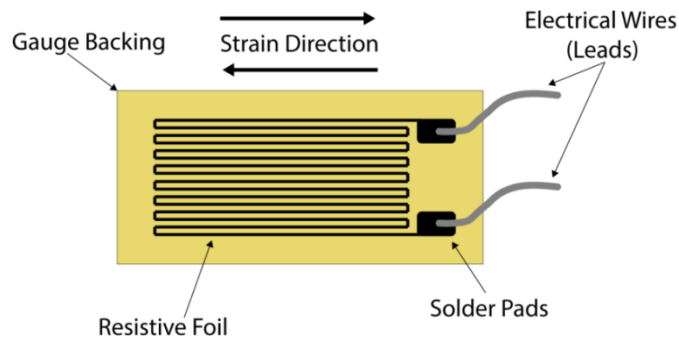
Measurement of pressure: strain gauge and piezoelectric type

I. Strain Gauges:

The strain gauge is a passive, resistive transducer which converts the mechanical elongation and compression into a resistance change. It can be used for measurement of force, torque, pressure, acceleration and many other parameters. The basic principle of operation of a strain gage is simple: when strain (force, pressure, tension, weight etc.) is applied to a thin metallic wire, its dimension changes, thus changing the resistance of the wire. This change in resistance takes place due to variation in length and cross sectional area of the gauge wire, when an

external force acts on it. Strain gauges are widely used in experimental stress analysis and diagnosis on machines and failure analysis.

Strain Gauge



A fundamental parameter of the strain gauge is its sensitivity to strain, expressed quantitatively as the gauge factor (GF). Gauge factor is defined as the ratio of fractional change in electrical resistance to the fractional change in length (strain):

$$GF = \frac{\Delta R/R}{\Delta L/L} = \frac{\Delta R/R}{\epsilon}$$

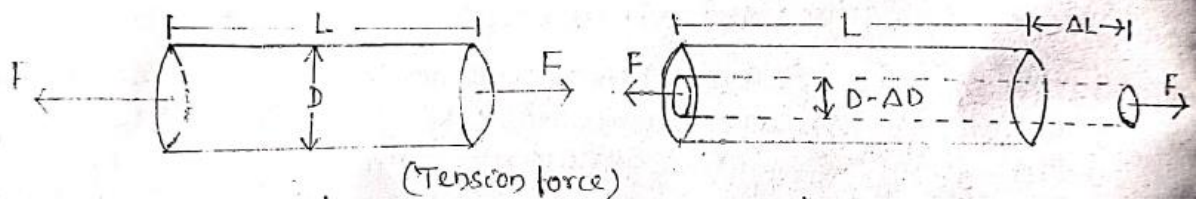
For elastic materials, stress-strain relationship is linear i.e. if stress is removed deformation will vanish.

Modulus of elasticity $E = \frac{\text{Stress}}{\text{Strain}}$. Stress has the unit of Newton per square meter while strain is unitless. When a member is subjected to simple uniaxial stress in the elastic range, the relation between lateral strain and axial strain ϵ_s given by the following equation.

$$\mu = -\frac{\epsilon_L}{\epsilon_a} \quad \mu = \text{poisson Ratio.}$$

A -ve strain refers to compression rather than elongation.

Derivation of Gauge Factor :-



Let us consider a strain gauge made of circular wire. The wire has the dimensions: Length = L , area = A , diameter = D before being strained. The material of the wire has a resistivity ρ .

∴ Resistance of unstrained gauge $R = \rho \frac{L}{A}$.
 Let a tensile stress S be applied to the wire. This produces a positive strain causing the length to increase and area to decrease as shown in above figure. Thus when the wire is strained there are changes in its dimensions. Let ΔL = change in length, ΔA = change in area, ΔD = change in diameter, and ΔR = change in resistance.

In order to find how ΔR depends upon the material physical quantity, the expression for R is differentiated w.r.t. stress S . Thus we get

$$\frac{dR}{dS} = \frac{\rho}{A} \cdot \frac{\partial L}{\partial S} - \frac{\rho L}{A^2} \frac{\partial A}{\partial S} + \frac{L}{A} \frac{\partial \rho}{\partial S} \quad \text{--- (1)}$$

Dividing through out by resistance. $R = \rho \frac{L}{A}$, we have

$$\frac{1}{R} \frac{dR}{dS} = \frac{1}{L} \frac{\partial L}{\partial S} - \frac{1}{A} \frac{\partial A}{\partial S} + \frac{1}{\rho} \frac{\partial \rho}{\partial S} \quad \text{--- (2)}$$

From this equation it is evident that percent change in resistance is due to

- (i) Percent change in length = $\frac{\Delta L}{L}$
- (ii) Percent change in area = $\frac{\Delta A}{A}$ and
- (iii) Percent change in resistivity = $\frac{\Delta \rho}{\rho}$.

Area $A = \frac{\pi}{4} D^2$ ∴ $\frac{\partial A}{\partial S} = 2 \cdot \frac{\pi}{4} D \cdot \frac{\partial D}{\partial S}$
 $\Rightarrow \frac{1}{A} \frac{\partial A}{\partial S} = \frac{(2\pi/4) D}{(\pi/4) D^2} \cdot \frac{\partial D}{\partial S} = \frac{2}{D} \cdot \frac{\partial D}{\partial S}$

Equation (2) can be written as:

$$\frac{1}{R} \cdot \frac{dR}{dS} = \frac{1}{L} \frac{\partial L}{\partial S} - \frac{2}{D} \frac{\partial D}{\partial S} + \frac{1}{\rho} \frac{\partial \rho}{\partial S}.$$

Now, Poisson's ratio

$$\nu = \frac{\text{lateral strain}}{\text{longitudinal strain}} = - \frac{\partial D/D}{\partial L/L}$$

$$\partial D/D = -\nu \times \partial L/L$$

$$\therefore \frac{1}{R} \frac{dR}{ds} = \frac{1}{L} \frac{\partial L}{\partial s} + \nu \frac{2}{L} \frac{\partial L}{\partial s} + \frac{1}{\rho} \frac{\partial \rho}{\partial s}$$

For small variation, the above relationship can be written as

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + 2\nu \frac{\Delta L}{L} + \frac{\Delta \rho}{\rho}$$

The Gauge factor is defined as the ratio of percent change in resistance to percent change in length.

$$\text{Gauge factor } G_f = \frac{\Delta R/R}{\Delta L/L}$$

$$\frac{\Delta R}{R} = G_f \frac{\Delta L}{L} = G_f \cdot \epsilon$$

Gauge factor can be written as:

$$= 1 + 2\nu + \frac{\Delta \rho / \rho}{\epsilon}$$

$\downarrow$
Resistance change due to change of length

$\downarrow$
Resistance change due to change in area.

$\downarrow$
Resistance change due to piezoresistive effect.

$$G_f = \frac{\Delta R}{R} / \frac{\Delta L}{L} = 1 + 2\nu + \frac{\Delta \rho / \rho}{\Delta L / L}$$

The strain is usually expressed in terms of microstrain
1 microstrain = $1 \mu\text{m/m}$.

If the change in the value of resistivity of a material when strained is neglected, the gauge factor is

$$G_f = 1 + 2\nu$$

This eqⁿ is valid only when piezoelectric effect.
i.e. change in resistivity due to strain is almost negligible. Normal gauge factor of the wire lies betⁿ
2 to 2.2

Strain Gauge Transducer Types:

Strain Gauge Transducer Types are three types, namely

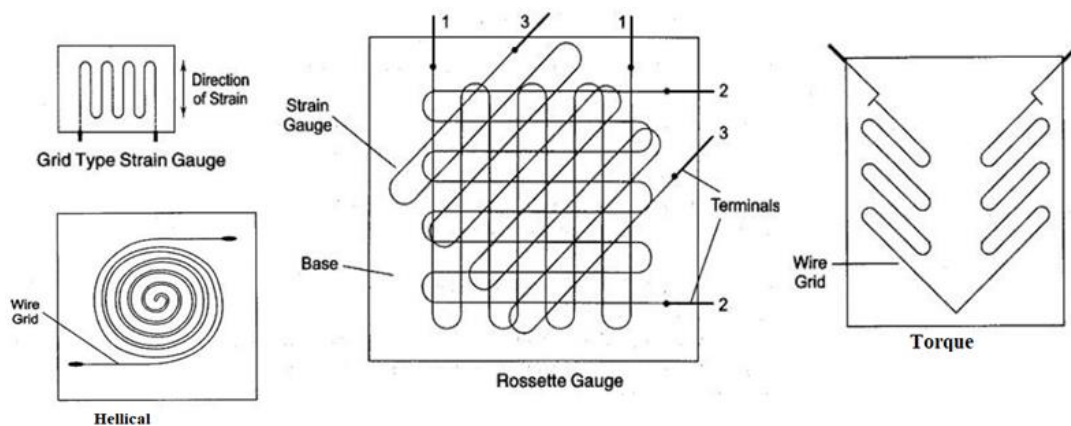
- Wire Strain Gauges
- Foil Strain Gauge
- Semiconductor Strain Gauge

Wire Strain Gauges: Wire Strain Gauges has three types namely: Grid type, Rossette type, Torque type, Helical type.

The grid arrangement of the wire element in a bonded strain gauge creates a problem not encountered in the use of unbonded strain gauges. To be useful as a strain gauge, the wire element must measure strain along one axis. Therefore complete and accurate analysis of strain in a rigid member is impossible, unless the direction and magnitude of stress are known.

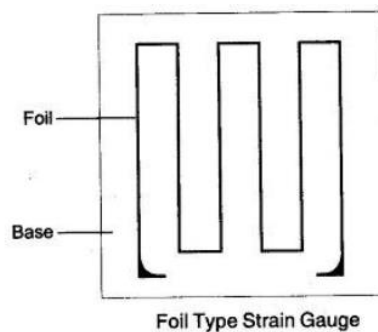
The 45° Rossette gauge is, in general, the most popular one. There are different shapes and sizes of Strain Gauge Transducer Types for various purposes.

The Strain Gauge Transducer Types should have a high value of gauge factor (a high value of gauge factor indicates a large change in resistance for particular strain, implying high sensitivity).



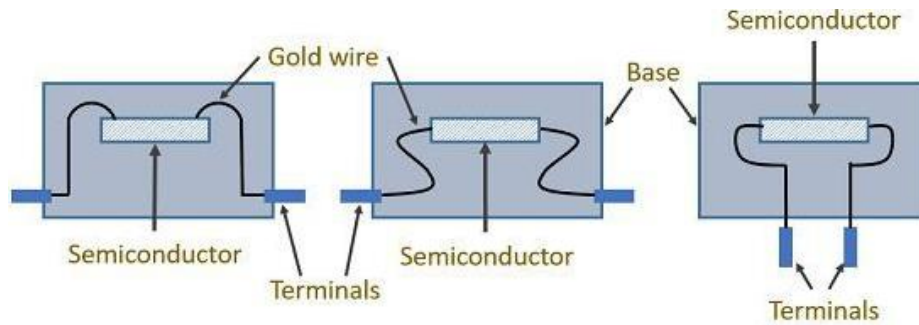
Foil Strain Gauge:

This class of strain gauges is an extension of the resistance wire strain gauge. The strain is sensed with the help of a metal foil. The metals and alloys used for the foil and wire are nichrome, constantan (Ni + Cu), isoelastic (Ni + Cr + Mo), nickel and platinum.



Semiconductor Strain Gauge:

Semiconductor strain gauges are used when a very high gauge factor is required. They have a gauge factor 50 times as high as wire strain gauges. The resistance of the semiconductor changes with change in applied strain. Semiconductor materials such as germanium and silicon are used as resistive materials. The gauge factor of this type of semiconductor strain gauge is $130 \pm 10\%$ for a unit of 350Ω , 1" long, 1/2" wide and 0.005" thick.



Advantages of Semiconductor Strain Gauge

- ✓ Semiconductor strain gauges have a high gauge factor of about + 130. This allows measurement of very small strains, of the order of 0.01 micro
- ✓ Hysteresis characteristics of semiconductor strain gauges are excellent, e. less than 0.05%.
- ✓ Semiconductor strain gauges can be very small in size, ranging in length from 0.7 to 7.0 mm.

Disadvantages of Semiconductor Strain Gauge

- ✓ They are very sensitive to changes in temperature.
- ✓ Linearity of semiconductor strain gauges is poor.
- ✓ They are more expensive.

Example 25'7. A resistance wire strain gauge uses a soft iron wire of small diameter. The gauge factor is +4'2. Neglecting the piezoresistive effects, calculate the Poisson's ratio.

Solution. The gauge factor is given by Eqn. 25'30, $G_f = 1 + 2\nu + \frac{\Delta \rho / \rho}{\epsilon}$

If piezoresistive effect is neglected, the gauge factor is given by Eqn. 25'32 as : $G_f = 1 + \nu$

$$\therefore \text{Poisson's ratio } \nu = \frac{G_f - 1}{2} = \frac{4'2 - 1}{2} = 1'6.$$

Example 25'8. A compressive force is applied to a structural member. The strain is 5 micro-strain. Two separate strain gauges are attached to the structural member, one is a nickel wire strain gauge having a gauge factor of -12'1 and the other is nichrome wire strain gauge having a gauge factor of 2. Calculate the value of resistance of the gauges after they are strained. The resistance of strain gauges before being strained is 120 Ω .

Solution. According to our convention, the tensile strain taken as positive while the compressive strain is taken as negative. Therefore, strain $\epsilon = -5 \times 10^{-6}$ (1 micro strain = 1 $\mu\text{m/m}$)

Now $\Delta R/R = G_f \epsilon$ (See Eqn. 25'29)

$$\therefore \text{Change in value of resistance of nickel wire strain gauge : } \Delta R = G_f \epsilon \times R \\ = (-12'1) \times (-5 \times 10^{-6}) \times 120 = 7'26 \times 10^{-3} \Omega = 7'26 \text{ m}\Omega.$$

Thus there is an increase of 7'26 m Ω in the value of resistance.

For nichrome, the change in the value of resistance is :

$$\Delta R = (2) \times (-5 \times 10^{-6}) \times 120 = -1'2 \times 10^{-3} \Omega = -1'2 \text{ m}\Omega.$$

Thus with compressive strain, the value of resistance gauge shows a decrease of 1'2 m Ω in the value of resistance.

Example 25.9. A resistance, wire strain gauge with a gauge factor of 2 is bonded to a steel structural member subjected to a stress of 100 MN/m^2 . The modulus of elasticity of steel is 200 GN/m^2 . Calculate the percentage change in the value of the gauge resistance due to the applied stress. Comment upon the results.

Solution. Strain $\epsilon = \frac{s}{E} = \frac{100 \times 10^6}{200 \times 10^9} = 500 \times 10^{-6}$. (500 microstrain).

We have : $\frac{\Delta R}{R} = G_f \epsilon = 2 \times 500 \times 10^{-6} = 0.001 = 0.1\%$.

$\therefore$ The change in resistance is only 0.1%.

II. Piezoelectric Sensors

- ✓ The Piezoelectric transducer is an electroacoustic transducer use for conversion of pressure or mechanical stress into an alternating electrical force.
- ✓ It is used for measuring the physical quantity like force, pressure, stress, etc., which is directly not possible to measure.
- ✓ The piezo transducer converts the physical quantity into an electrical voltage which is easily measured by analogue and digital meter.
- ✓ The piezoelectric transducer uses the piezoelectric material which has a special property, i.e. the material induces voltage when the pressure or stress applied to it. The material which shows such property is known as the electro-resistive element.
- ✓ The word piezoelectric means the electricity produces by the pressure. The Quartz is the examples of the natural piezoelectric crystals, whereas the Rochelle salts, ammonium dehydration, phosphate, lithium sulphate, dipotassium tartrate are the examples of the man made crystals. The ceramic material is also used for piezoelectric transducer.
- ✓ The ceramic material does not have the piezoelectric property. The property is developed on it by special polarizing treatment. The ceramic material has several advantages. It is available in different shapes and sizes. The material has the capability of working at low voltages, and also it can operate at the temperature more than 3000°C

Piezoelectric Effect

The EMF develops because of the displacement of the charges. The effect is changeable, i.e. if the varying potential applies to a piezoelectric transducer, it will change the dimension of the material or deform it. This effect is known as the piezoelectric effect.

The pressure is applied to the crystals with the help of the force summing devices for examples the stress is applied through mechanical pressure gauges and pressure sensors, etc. The deformation induces the EMF which determines the value of applied pressure.

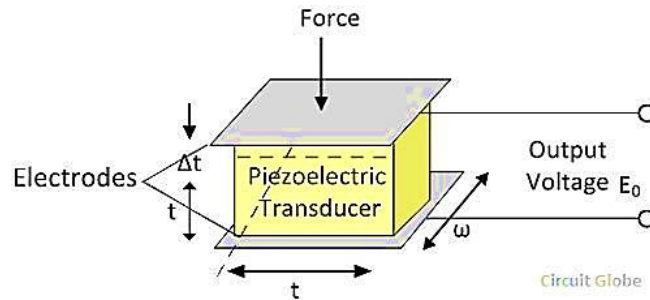
Piezoelectric Materials

Natural: Quartz, Rochellesalt, Topaz, Tourmaline-group minerals, and some organic substances as silk, wood, enamel, bone, hair, rubber, dentin.

Artificial: Polyvinylidenedifluoride, PVDF or PVF2, Bariumtitanate, Lead titanate, Lead zirconate titanate (PZT), Potassium niobate, Lithium niobate, Lithium tantalate, and other lead-free piezo electric ceramics.

Theory of Piezo-Electric Transducer

A piezoelectric crystal is shown in the figure below.



The polarity of the charge depends on the direction of the applied forces.

$$\text{Charge } Q = d \times F \text{ Coulomb}$$

Where, d – charge sensitivity of the crystals, F – applied force in Newton
The force changes the thickness of the crystals.

$$F = \frac{AE}{t} \Delta t \text{ Newton}$$

Where A – area of crystals in meter square

t – the thickness of crystals in meter, E – Young's modulus N/m^2

The young modulus is,

$$E = \frac{\text{stress}}{\text{strain}} = \left(\frac{F}{A} \right) \cdot \frac{1}{\Delta t/t}$$

$$E = \frac{Ft}{A\Delta t} \text{ N/m}^2$$

$$A = \omega l$$

where ω – width of crystals in meter, l – the length of crystals in meter

On substituting the value of force in the equation of charge, we get

$$Q = dAE \left(\frac{\Delta t}{t} \right)$$

The output voltage is obtained because of the electrode charges.

$$E_o = \frac{Q}{C_p} = \frac{dF}{\epsilon_r \epsilon_0 A/t}$$

$$E_o = \frac{d}{\epsilon_r \epsilon_0} tP$$

$$E_o = gtP$$

$$g = \frac{d}{\epsilon_r \epsilon_0}$$

The g is the voltage sensitivity of the crystals.

$$g = \frac{E_o}{tP} = \frac{E_o/t}{P}$$

Where E_o – electric field strength, V/m

The voltage sensitivity of the crystals is expressed by the ratio of the electric field intensity and pressure.

When the mechanical deformation occurs in the crystals, it generates charges. And this charge develops the voltages across the electrodes.

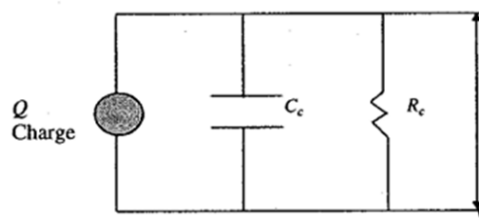
The Piezoelectric crystal is direction sensitive. The polarity of the voltage depends on the direction of the force which is either tensile or compressive. The magnitude and the polarity of the charges depend on the magnitude and the direction of the applied force.

Equivalent Circuit of a piezoelectric Transducer

The equivalent circuit can be derived from the electrical and mechanical parameters of the transducer

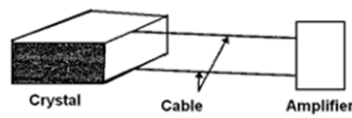
$$V = \frac{Q}{C} = \frac{dF}{C_c}$$

The charge generated Q is across the capacitance C_c and its leakage resistance is R_c



$$V = \frac{Q}{C} = \frac{d \cdot F}{C_c}$$

When the piezoelectric crystal is coupled with leads and cables as well as a readout device, the voltage depends not only on the element but also on the capacitance of cables, charge amplifier and oscilloscope display. The total capacitance is



$$C = C_c + C_{\text{cable}} + C_{\text{display}}$$

TC amplifier display

Advantages

- ✓ These are active transducers. It means that they don't need external power and force for operation and are self-producing.
- ✓ The high-frequency reaction of these piezoelectric transducers makes them a perfect choice for different applications because some parameters in these applications change too quickly.
- ✓ They are easy to handle and utilize due to their small dimensions.

Disadvantages

- ✓ Environmental conditions like temperature variations can influence the behavior of the piezoelectric transducers.
- ✓ They are not suitable for measurement in static applications. Piezoelectric Transducers can only measure dynamic pressure and are useless to measure static parameters.
- ✓ The output voltage of Piezoelectric Transducers is occasionally low in some applications. So, some external amplifiers must attach to their circuit.
- ✓ It is very hard to give the desired strength and shape to piezoelectric materials. Therefore, piezoelectric transducers always utilize in the same shape and design in different applications.

Applications

- ✓ Due to its excellent frequency response, it is normally used as an accelerometer, where the output is in the order of (1-30) mV per gravity of acceleration.
- ✓ The device is usually designed for use as a pre-tensional bolt so that both tensional and compression force measurements can be made.
- ✓ It can be used for measuring force, pressure and displacement in terms of voltage
- ✓ The piezoelectric material has high stability and hence it is used for stabilizing the electronic oscillator.
- ✓ The ultrasonic generators use the piezoelectric material. This generator is used in SONAR for underwater detection and in industrial apparatus for cleaning.
- ✓ It is used in microphones and speakers for converting the electric signal into sound.
- ✓ The piezoelectric material is used in electric lighter.

Example 25.31. A barium titanate pickup has the dimensions of $5 \text{ mm} \times 5 \text{ mm} \times 1.25 \text{ mm}$. The force acting on it is 5 N. The charge sensitivity of barium titanate is 150 pC/N and its permittivity is $12.5 \times 10^{-9} \text{ F/m}$. If the Modulus of elasticity of barium titanate is $12 \times 10^6 \text{ N/m}^2$, calculate the strain. Also calculate the charge and the capacitance

Solution. Area of plates $A = w \times l = 5 \times 5 \text{ mm}^2 = 25 \times 10^{-6} \text{ m}^2$.

Pressure $P = \frac{F}{A} = \frac{5}{25 \times 10^{-6}} \text{ N/m}^2 = 0.2 \text{ MN/m}^2$.

Voltage sensitivity $g = \frac{d}{\epsilon_0 \epsilon_r} = \frac{150 \times 10^{-2}}{12.5 \times 10^{-9}} = 12 \times 10^{-3} \text{ Vm/N}$.

Voltage generated $E_0 = g \times P = 12 \times 10^{-3} \times 0.2 \times 10^6 = 3 \text{ V}$.

Strain $\epsilon = \frac{\Delta l}{l} = \frac{\text{stress}}{\text{Young's modulus}} = \frac{0.2 \times 10^6}{12 \times 10^6} = 0.0167 \text{ m/m}$.

Charge $Q = dF = 150 \times 10^{-12} \times 5 \text{ C} = 750 \text{ pC}$.

Capacitance $C_p = \frac{Q}{E_0} = \frac{750 \times 10^{-12}}{3} \text{ F} = 250 \text{ pF}$.

Ultrasonic transducer:

An ultrasonic transducer is a device that converts electrical energy into ultrasonic waves (sound waves with frequency greater than 20 kHz) and, in some cases, converts reflected ultrasonic waves back into electrical signals. It calculates distance by emitting sound waves and collecting their echoes. The ultrasonic transducer is one type of sound-related sensor. These transducers send the electrical signals to the object and once the signal strikes the object then it reverts to the transducer. In this process, this transducer measures the distance of the object. It can measure objects from up to 4.5 meters away, which makes it a versatile instrument for correctly measuring both short and long distances without making contact with the target object.

Construction:

An ultrasonic transducer consists of the following parts:

1. Piezoelectric crystal – generates ultrasonic vibrations
2. Electrodes – supply alternating electrical voltage

3. Backing material – absorbs unwanted vibrations
4. Matching layer – improves transmission efficiency
5. Casing – provides protection and insulation

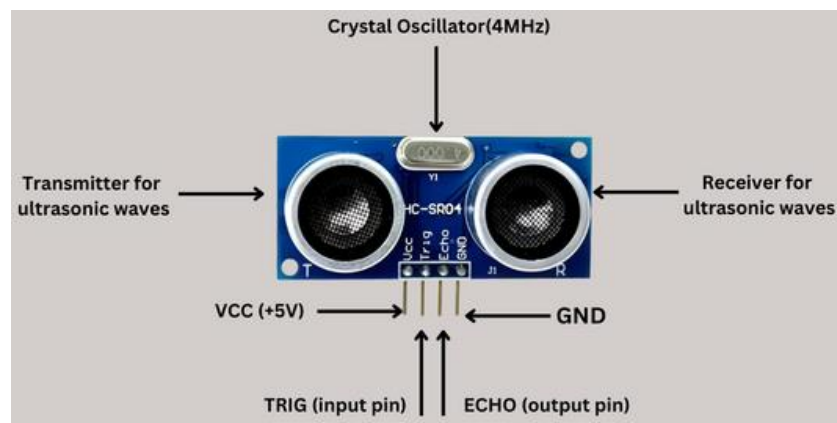
There are four pins in the sensor,

Vcc: power supply +5 V

Gnd: Common ground

Trigger pin: To start the sensor

Eco pin: Receive the signal



Ultrasonic Transducer Working Principle

When an electrical signal is applied to this transducer, it vibrates around the specific frequency range and generates a sound wave. These sound waves travel and whenever any obstacle comes, these sound waves will reflect the transducer inform of echo. And at the end of the transducer, this echo converts into an electrical signal. Here, the transducer calculates the time interval between the sending of the sound wave to the receiving the echo signal. The ultrasonic sensor sends the ultrasonic pulse at 40 kHz which travels through the air. These transducers are better than the infrared sensors because these ultrasonic transducer/sensors are not affected by the smoke, black materials, etc. Ultrasonic sensors exhibit excellence in suppressing background interference.

Ultrasonic transducers are mainly used for finding the distance by using ultrasonic waves. The distance can be measured by the following formula.

$D = \frac{1}{2} \times T \times C$, Here, D indicates the distance, T - indicates the time difference between sending and reception of ultrasonic waves, C is indicating the sonic speed.

In summary:

- An AC voltage of ultrasonic frequency is applied across the electrodes.
- The piezoelectric crystal vibrates due to the inverse piezoelectric effect.
- These vibrations produce ultrasonic waves.
- When the reflected ultrasonic waves strike the crystal, electrical signals are produced due to the direct piezoelectric effect.

Principle: Ultrasonic transducers operate based on the piezoelectric effect.

- Inverse piezoelectric effect: When an alternating voltage is applied to a piezoelectric crystal, it vibrates and produces ultrasonic waves.
- Direct piezoelectric effect: When ultrasonic waves strike the crystal, mechanical vibrations generate an electrical signal.

Common piezoelectric materials include:

- Quartz
- Lead Zirconate Titanate (PZT)
- Rochelle salt

Ultrasonic Transducer Types

There are various types of ultrasonic transducers available based on factors like piezoelectric crystal arrangement, footprint, and frequency. They are

- ✓ *Linear Ultrasonic Transducers* – In this type of transducers, piezoelectric crystal arrangement is linear.
- ✓ *Standard Ultrasonic Transducers* – This type is also called as convex transducers. In this type, the piezoelectric crystal is in a curvy form. For in-depth examinations these are preferable.
- ✓ *Phased Array Ultrasonic Transducers* – Phased array transducers have a small footprint and low frequency. (its center frequency is 2 MHz – 7 MHz)
- ✓ *Piezoelectric Ultrasonic Transducers* – widely used, high efficiency
- ✓ *Magnetostrictive Ultrasonic Transducers* – based on magnetostriction
- ✓ *Capacitive Micromachined Ultrasonic Transducers (CMUTs)* – used in advanced systems

Applications

- ✓ Medical ultrasound imaging
- ✓ SONAR systems
- ✓ Non-destructive testing (NDT)
- ✓ Distance and level measurement
- ✓ Industrial cleaning
- ✓ Flow measurement

Advantages:

- ✓ High accuracy
- ✓ Non-invasive and non-destructive
- ✓ Compact and reliable

- ✓ Suitable for solids and liquids

Disadvantages

- ✓ Ultrasonic transducers are sensitive to temperature variation. This temperature variation may change the ultrasonic reaction.
- ✓ It will face problems while reading the reflections from small objects, thin and soft objects.
- ✓ Limited penetration in air
- ✓ Requires proper coupling medium

Linear variable differential transformer

I. Linear Variable Differential Transformer (LVDT)

- ✓ LVDT is an inductive type passive transducer.
- ✓ It measures force in terms of displacement of ferromagnetic core of a transformer.
- ✓ It converts translational or linear displacement into electrical voltage.
- ✓ It is also known as Linear Variable Differential Transducer.

Principle

- ✓ It is based on the principle of electro-magnetic induction

Construction

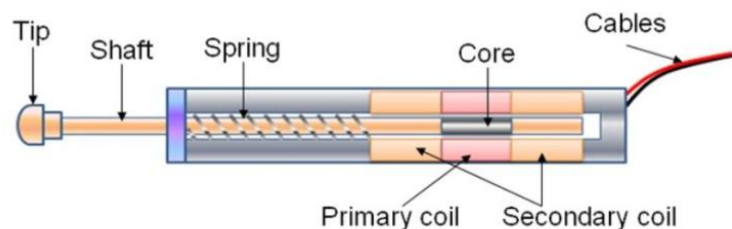


Figure 2.2.6 Construction of LVDT sensor

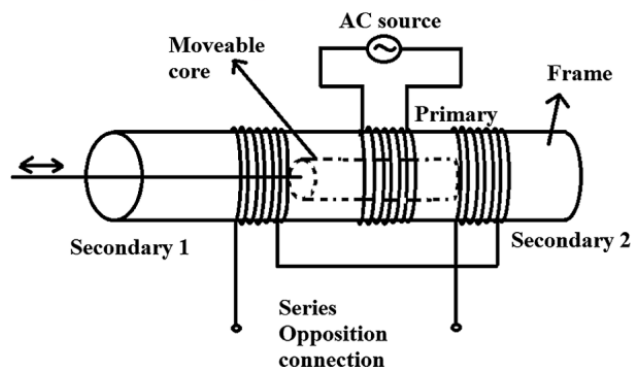
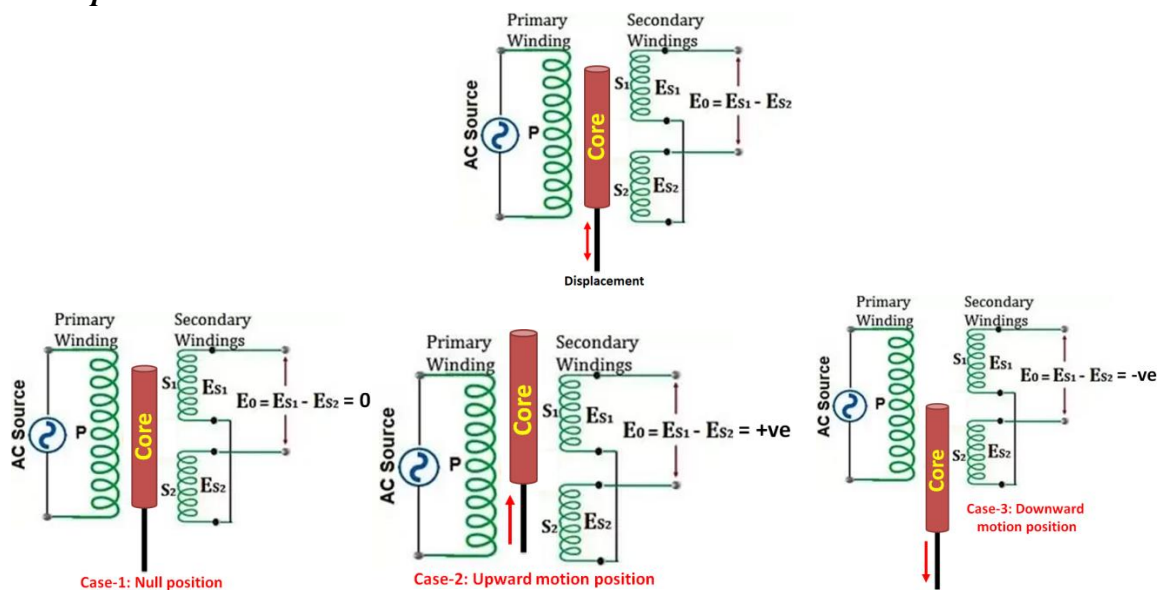


Fig. Diagram of Linear Variable Differential Transformer (LVDT)

- ✓ LVDT consist of cylindrical transformer where it is surrounded by one primary winding in the centre of the former and two secondary windings at the sides. The central coil is primary coil and the other two are secondary coils.
- ✓ The numbers of turns in both the secondary windings are equal, but they are opposite to each other. Secondary coils are connected in series in such a way that their outputs oppose each other.
- ✓ The primary winding is connected to the ac source.
- ✓ A movable soft iron core slides within hollow former and therefore affects magnetic coupling between primary and two secondary. A magnetic core attached to the element of which displacement is to be monitored is placed inside the insulated tube.

Operation:



- ✓ Due to an alternating voltage input to the primary coil, alternating electro-magnetic forces (emfs) are generated in secondary coils.
- ✓ When the iron core lies at the centre of both secondary, the output differential voltage remains unaffected and have zero magnitude.
- ✓ When the core moves towards secondary-1, it induces more emf across it and less emf across secondary-2. Let's assume that it is positive displacement. Due to more flux links with the secondary-1 than secondary-2.
- ✓ When the core moves towards secondary-2, it induces more emf across it and less emf across secondary-1. Let's assume that it is negative displacement. Due to more flux links with the secondary-2 than secondary-1.
- ✓ The output differential voltage is proportional to the displacement of the iron core.

Output Voltage:

Differential output voltage $E_0 = E_{S1} - E_{S2}$

Now the voltage induced in the secondary coil is given by

$$E_{s1} = M_1 \frac{di_p}{dt}$$

$$E_{s2} = M_2 \frac{di_p}{dt}$$

Where M_1 and M_2 are the respective mutual inductance.

The net secondary voltage is given by

$$E_o = E_{s1} - E_{s2} = (M_1 - M_2) \frac{di_p}{dt}$$

Here both M_1 and M_2 are linear function of displacement x . So $M_1 - M_2$ will also be a linear function of x . Hence we may write

$$M_1 - M_2 = Kx$$

$$\text{Then } E_o = E = Kx \frac{di_p}{dt} = K \frac{di_p}{dt} X$$

So $E_o \propto x$.

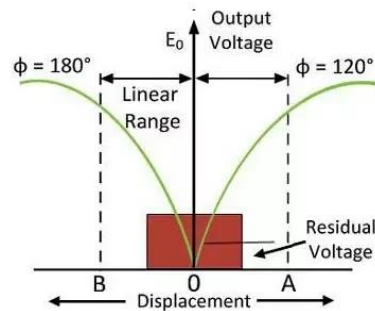


Fig. Variation in output voltage with linear displacement for an LVDT

- ✓ **CASE I :** When the core is at its normal (NULL) position, the flux linking with both the secondary windings is equal and hence equal emfs are induced in them. Thus at null position $E_{s1} = E_{s2}$. Since the output voltage of the transducer is the difference of the two voltages, the output voltage E_o is zero at null position.
- ✓ **CASE II :** Now if the core is moved to the left of the NULL position, more flux links with winding S1 and less with winding S2. Accordingly output voltage E_{s1} , of the secondary winding S1, is more than E_{s2} , the output voltage of secondary winding S2. The magnitude of output voltage is, thus, $E_{s1} - E_{s2}$ and the output voltage is in phase with E_{s1} i.e. the output voltage of secondary winding S1.
- ✓ **CASE III:** Similarly, if the core is moved to the right of the null position, the flux linking with winding S2 becomes larger than that linking with winding S1. This results in E_{s2} becoming larger than E_{s1} . The output voltage in this case is $E_o = E_{s2} - E_{s1}$ and is in phase with E_{s2} i.e. the output voltage of secondary winding S2.

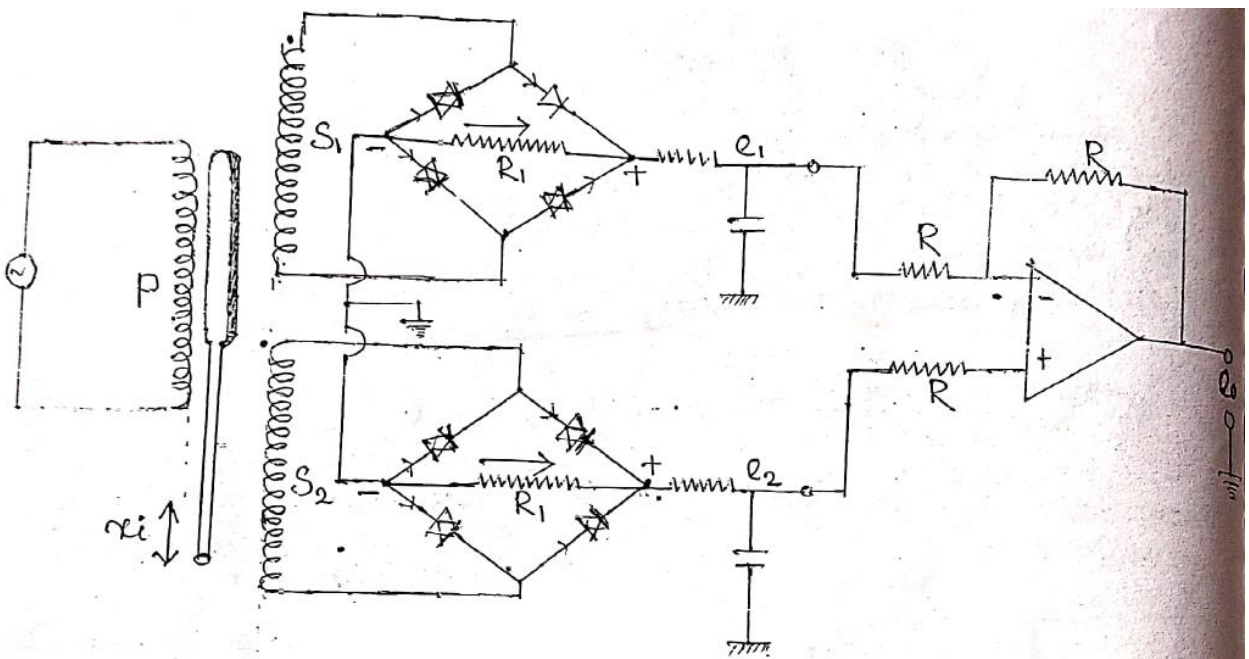
180° phase shift $\rightarrow X_c$. The cause of residual voltages are
 (c) If the two secondaries are not identical i.e. they may not have same no. of turns, they may not be same dimensions.

(e) The stray capacitance coupling betⁿ the primary and the secondary. Also differing ear capacitance produce residual voltage.

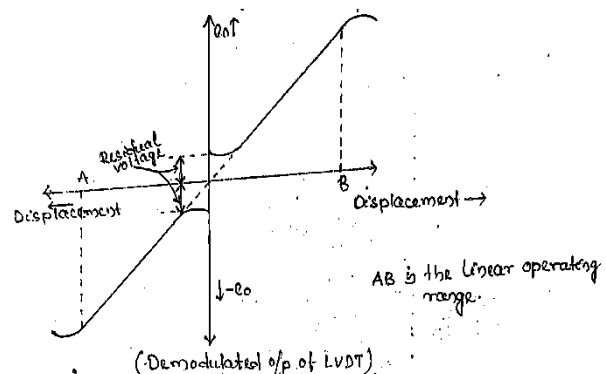
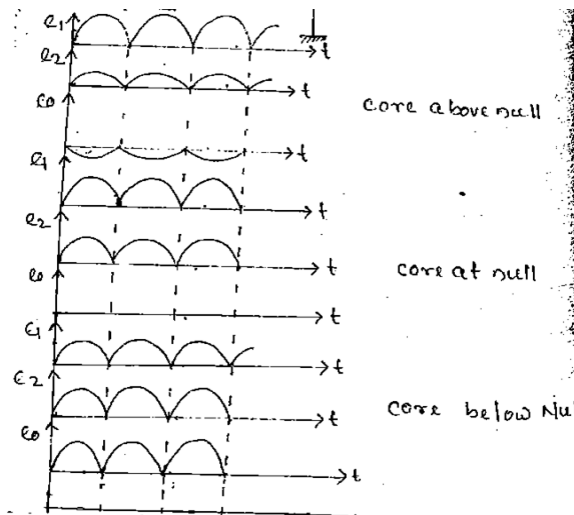
(e) Harmonics in the excitation voltage. That is the secondaries may not be balanced at null position for the harmonics present in the excitation but may be balanced for the fundamental frequency of the excitation.

(e) Stray magnetic coupling may also produce residual voltage. This may be overcome by the use of magnetic shield.

Demodulation Network:



Demodulation ckt consists of two full wave rectifier bridge connected to the secondary terminals. The o/p of the two bridges are connected to a differential amplifier which gives a polarity output indicating the position of the core w.r.t null. The amplitude of the amplifier o/p gives the indication of the magnitude of the c/p displacement. The ckt shown below.



Advantages

- ✓ High range (1.25 mm to 250 mm)
- ✓ No frictional losses
- ✓ High input and high sensitivity
- ✓ Low hysteresis
- ✓ Low power consumption
- ✓ Direct conversion to electrical signals

Dis-advantages

- ✓ LVDT is sensitive to stray magnetic fields so they always require a setup to protect them from stray magnetic fields.
- ✓ They are affected by vibrations and temperature.

Applications

- ✓ It is used where displacements ranging from fraction of mm to few cm are to be measured.
- ✓ It act as primary transducer.
- ✓ They can also act as secondary transducer. E.g. the bourdon tube which acts as primary transducer and convert pressure into linear displacement then LVDT converts it into electrical signal.

Capacitance type transducer

Capacitive Transducers:

II. Variable Distance

III. Variable Area

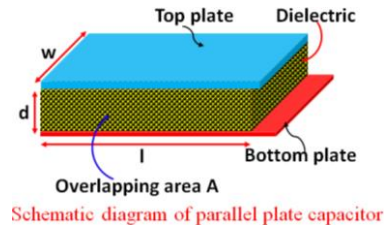
IV. Variable Dielectric Type Capacitive Sensors

Capacitive sensor is of non-contact type sensor and is primarily used to measure the linear displacements from few milli meters to hundreds of milli meters.

It comprises of three plates, with the upper pair forming one capacitor and the lower pair another.

❖ The linear displacement might take in two forms:

- ✓ one of the plates is moved by the displacement so that the plate separation changes
- ✓ Area of overlap changes due to the displacement.



Construction and working:

The capacitive transducer comprises of two parallel metal plates that are separated by the material such as air, which is called as the dielectric material. In the typical capacitor, the distance between the two plates is kept varying. In the instruments using capacitance transducers the value of the capacitance changes due to the change in the value of the input quantity that is to be measured. This change in capacitance can be measured easily and it is calibrated against the input quantity, thus the value of the input quantity can be measured directly.

The principle of operation of capacitive transducers are based on the following expression,

The capacitance C of a parallel plate capacitor is given by $C = \epsilon_r * \epsilon_0 (A/d)$

Where, ϵ_r is the relative permittivity of the dielectric between the plates,

ϵ_0 permittivity of free space,

A area of overlap between two plates and

d the plate separation.

Depending on the parameter that changes for the capacitive transducers, they are of three types as mentioned below:

1) Changing Dielectric Constant type of Capacitive Transducers

In this type of capacitive transducer the dielectric material between the two plates changes, due to which the capacitance of the transducer also changes. When there is a change in the input quantity to be measured, the value of the dielectric constant also changes providing a change in the capacitance of the capacitive transducer. This capacitance, calibrated against the input quantity, directly gives the value of the quantity to be measured.

This principle is used for measurement of level in the hydrogen container, where the change in level of hydrogen between the two plates results in change of the dielectric constant of the

capacitance transducer. Apart from level, this principle can also be used for measurement of humidity and moisture content of the air.

2) Changing Area of the Plates of Capacitive Transducers

The capacitance of the variable capacitance transducer also changes with the area of the two plates. This principle is used in the torque meter, used for measurement of the torque on the shaft. This comprises of the sleeve that has teeth cut axially and the matching shaft that has similar teeth at its periphery.

3) Changing Distance between the Plates of Capacitive Transducers

In these capacitive transducers the distance between the plates is variable, while the area of the plates and the dielectric constant remain constant. This is the most commonly used type of variable capacitance transducer.

For measurement of the displacement of the object, one plate of the capacitance transducer is kept fixed, while the other is connected to the object. When the object moves, the plate of the capacitance transducer also moves, this results in change in distance between the two plates and the change in the capacitance. The changed capacitance is measured easily and it calibrated against the input quantity, which is displacement. This principle can also be used to measure pressure, velocity, acceleration etc.

Mathematical calculation & necessary equation:

Principle operation based on the equation , $c = \frac{\epsilon a}{d}$, $c = \frac{\epsilon_0 \epsilon_r a}{d}$

a -overlapping plate area

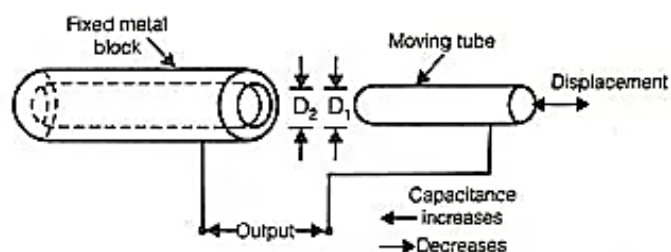
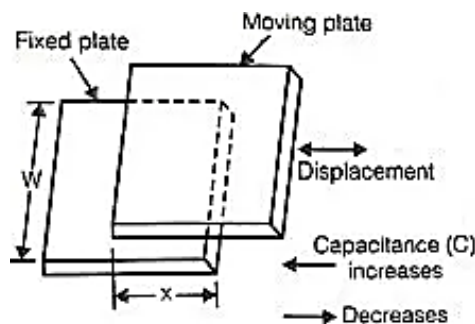
d -distance between the plates

$\epsilon_0 \epsilon_r$ - permittivity of the medium present between the plate

Capacitance transducer work on the principle of-

- (1) Change in overlapping area (a).
- (2) Change in distance between the plates (d).
- (3) Change in dielectric medium (ϵ).

(1) Change in overlapping area (a)



x- length of overlapping plate
w -width of the overlapping area
d –distance between them

$$\text{therefore , capacitance- } c = \frac{\epsilon a}{d} = \frac{\epsilon wx}{d}$$

$$\text{Sensitivity , } s = \frac{\partial c}{\partial x} = \frac{\epsilon w}{d}$$

Sensitivity for fractional change in capacitance-

$$s' = \frac{\partial c}{c \partial x} = \frac{1}{x}$$

For cylindrical capacitor—

x —length of overlapping region
d₂ – inner diameter of outer cylinder
d₁ – outer diameter of inner cylinder

$$\text{therefore , capacitance } c = \frac{2\pi \epsilon x}{\log_e d_2/d_1};$$

$$\text{Sensitivity } s = \frac{\partial c}{\partial x} = \frac{2\pi \epsilon}{\log_e d_2/d_1}$$

For circular capacitor—

Here the angular rotation of second plate changes the overlapping area. The capacitance is maximum when they are completely overlapping i.e. $\theta=180$

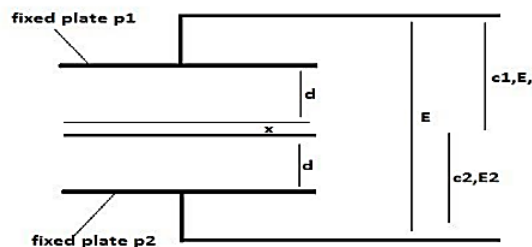
$$\text{Therefore, } c_{\max} = \frac{\epsilon a}{d} = \frac{\epsilon \pi r^2}{2d}$$

$$\text{And capacitance at angle } \theta, c_{\theta} = \frac{\epsilon \theta r^2}{2d}$$

$$\text{Sensitivity } s = \frac{\partial c}{\partial \theta} = \frac{\epsilon r^2}{2d}$$

(2) Change in distance between the plate (d)—

Differential arrangement:



When the moveable plate is not moving in either direction, initially both the capacitor having same value.

$$c_1 = \frac{\epsilon a}{d}, c_2 = \frac{\epsilon a}{d}$$

As, $C_1 = C_2$ then $E_1 = E_2$

$$\text{voltage across the } C_1 \text{ is-- } E_1 = \frac{Ec_2}{(c_1 + c_2)} = \frac{E}{2} \text{ and } E_2 = \frac{Ec_1}{(c_1 + c_2)} = \frac{E}{2}$$

therefore differential output, $\Delta E = E_1 - E_2 = 0$

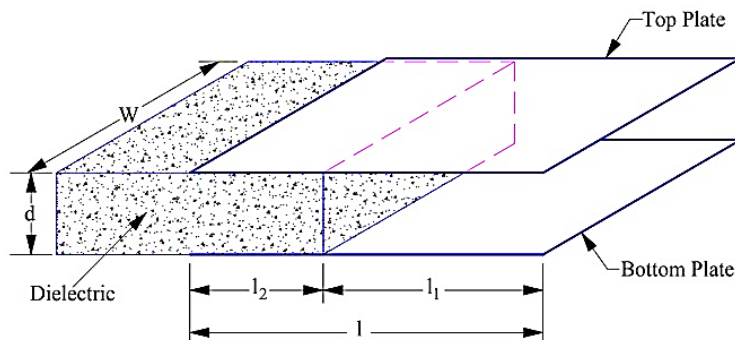
now moveable plate goes upward for a distance x , $c_1 = \frac{\epsilon a}{d-x}$ and $c_2 = \frac{\epsilon a}{d+x}$

$$\text{therefore, } E_1 = \frac{d-x}{2d} E \text{ and } E_2 = \frac{d+x}{2d} E$$

differential output voltage $\Delta E = E_2 - E_1 = \frac{Ex}{d}$, so, output voltage varies linearly with

displacement x . hence sensitivity, $s = \frac{\Delta E}{\Delta x} = \frac{E}{d}$

(3) Variation of di-electric constant for measurement of displacement—



For this arrangement initial capacitance is— $c = \epsilon_0 \frac{wl_1}{d} + \epsilon_0 \epsilon_r \frac{wl_2}{d}$

Now consider the capacitor moves distance x along the direction indicated. so change of capacitor is Δc

$$\text{So, } c + \Delta c = \frac{\epsilon_0 w}{d} (l_1 - x) + \frac{\epsilon_0 \epsilon_r w}{d} (l_2 + x)$$

$$= \epsilon_0 \frac{w}{d} [l_1 - x + \epsilon_r (l_2 + x)]$$

$$= \epsilon_0 \frac{w}{d} (l_1 + \epsilon_r l_2) + \epsilon_0 \frac{w}{d} (\epsilon_r - 1) x$$

$$= c + \epsilon_0 \frac{w}{d} (\epsilon_r - 1) x$$

Therefore change in capacitance-- $\Delta c = \epsilon_0 \frac{w}{d} (\epsilon_r - 1) x$

Here also change in capacitance proportional to displacement.

Advantages of Capacitive Transducers:

- ✓ They require extremely small forces to operate them and hence are very useful for use in small systems.
- ✓ They are extremely sensitive.
- ✓ They have a good frequency response.

- ✓ They have a high input impedance and therefore the loading effects are minimum.
- ✓ A resolution of the order of 2.5×10^{-3} mm can be obtained with these transducers.
- ✓ The capacitive transducers can be used for applications where stray magnetic fields render the inductive transducers useless.
- ✓ The force requirement of capacitive transducers is very small and therefore they require small power to operate them.

Disadvantages of Capacitive Transducers:

- ✓ The metallic parts of the capacitive transducers must be insulated from each other. In order to reduce the effects of stray capacitances, the frames must be earthed.
- ✓ The capacitive transducers show non-linear behaviour many a times on account of edge effects. Therefore guard rings must be used to eliminate this effect. Guard rings are also a must in order to eliminate the effect of stray electric fields, especially when the transducers have a low value of capacitance of the order of pF.
- ✓ The output impedance of capacitive transducers tends to be high on account of their small capacitance value.
- ✓ The cable connecting the transducer to the measuring point is also a source of error. The cable may be source of loading resulting loss of sensitivity. Also loading makes the low frequency response poor.

Applications of capacitive transducers:-

1. Feed hopper level monitoring
2. Small vessel pump control
3. Grease level monitoring
4. Level control of liquids
5. Metrology applications
 - ✓ to measure shape errors in the part being produced
 - ✓ to analyze and optimize the rotation of spindles in various machine tools such as surface grinders, lathes, milling machines, and air bearing spindles by measuring errors in the machine tools themselves
6. Assembly line testing
 - ✓ To test assembled parts for uniformity, thickness or other design features
 - ✓ To detect the presence or absence of a certain component, such as glue etc.

Numerical with Solution:

- A capacitive transducer with its plate separation 0.05mm under static condition has capacitance 5×10^{-12} F. find the displacement which cause change of capacitance of 0.75×10^{-12} F.

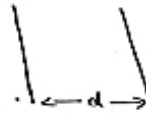
Soln:-

$$d = 0.05 \text{ mm}$$

$$C = 5 \times 10^{-12} \text{ F}$$

$$\Delta C = 0.75 \times 10^{-12} \text{ F}$$

$$\Delta d = ?$$



$$C = \frac{\epsilon A}{d}$$

Here, d is variable

$$Cd = \epsilon A = 0.25 \times 10^{-15}$$

$$\Delta C = \frac{\epsilon A}{\Delta d}$$

$$\Delta d = \frac{\epsilon A}{\Delta C}$$

$$\Rightarrow \frac{0.25 \times 10^{-15}}{0.75 \times 10^{-12}}$$

$$\Delta d \geq 0.33 \text{ mm}$$

- A capacitive transducer of two parallel plates of overlapping area = $5 \times 10^{-4} \text{ m}^2$ is immersed in water, capacitance is found to be 950pF. Calculate the separation between plate and sensitivity with respect to distance. Given relative permittivity of water is 81.

Solution:-

$$A_{\text{req}} = 5 \times 10^{-4} \text{ m}^2, C = 950 \text{ pF}$$

$$\epsilon_r = 81, \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$d = ?$$

$$\textcircled{1} C = \frac{\epsilon A}{d}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$d = \frac{8.854 \times 10^{-12} \times 81 \times 5 \times 10^{-4}}{950 \times 10^{-12}}$$

$$\boxed{d = 0.03775 \text{ m}}$$

$$\textcircled{2} \text{ Sensitivity, } \frac{\partial C}{\partial d} = ?$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$\frac{\partial C}{\partial d} = - \frac{\epsilon_0 \epsilon_r A}{d^2}$$

$$= \frac{8.854 \times 10^{-12} \times 81 \times 5 \times 10^{-4}}{(0.03775)^2}$$

$$\boxed{S \Rightarrow 0.025 \times 10^{-8} \text{ F/m}}$$

Proximity sensor

"Proximity Sensor" includes all sensors that perform non-contact detection in comparison to sensors, such as limit switches, that detect objects by physically contacting them. Proximity Sensors convert information on the movement or presence of an object into an electrical signal. There are three types of detection systems that do this conversion:

- ✓ Systems that use the eddy currents that are generated in metallic sensing objects by electromagnetic induction,
- ✓ Systems that detect changes in electrical capacity when approaching the sensing object,
- ✓ Systems that use magnets and reed switches

Types of Proximity sensor: There are four types of proximity sensor

1. Inductive proximity sensors
2. Capacitive proximity sensors.
3. Magnetic proximity sensors.
4. Optical proximity sensors

1. Inductive Proximity Sensors

The sensor incorporates an electromagnetic coil which is used to detect the presence of a conductive metal object. The sensor will ignore the presence of an object if it is not metal, Figure 1.1. This type of sensor consists mainly of four elements: coil, oscillator, trigger circuit, and an output, Figure 1.2. Inductive proximity sensors are designed to generate an electromagnetic field. When a metal object enters this field, surface currents, known as eddy currents, are induced in the metal object. These eddy currents drain energy from the electromagnetic field (causes a load on the sensor) resulting in a loss of energy in the oscillator circuit and, consequently, a reduction in the amplitude of oscillation. The trigger circuit detects this change and generates a signal to switch the output ON or OFF. When the object leaves the electromagnetic field area, the field regenerates and the sensor returns to its normal state. The oscillator is not affected by moisture and dusty/dirty environments.

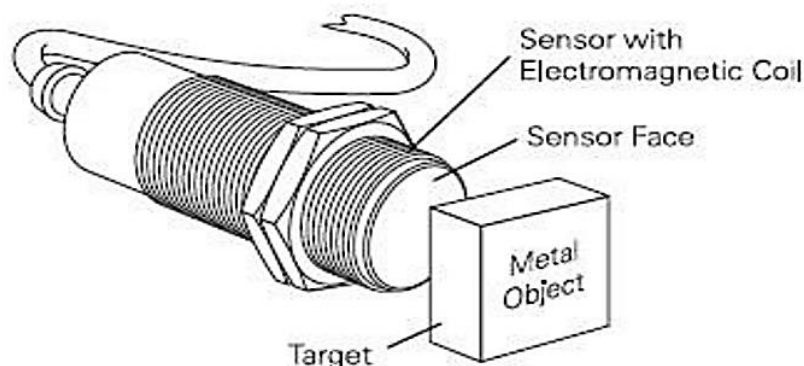


Figure 1.1: Inductive Proximity Sensor

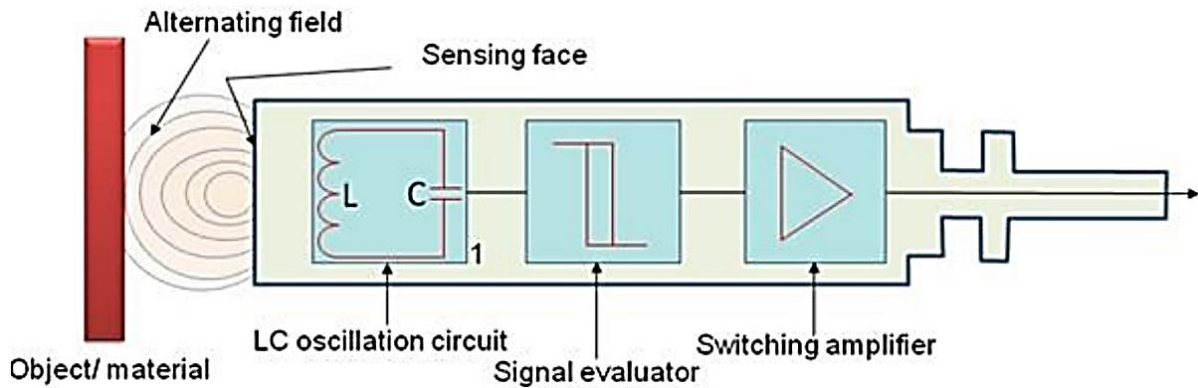


Figure 1.2: Construction of an Inductive Proximity Sensor

The operating distance of an inductive proximity sensor varies for each target and application. The ability of a sensor to detect a target is determined by the material of the metal target, its size, and its shape.

The advantages of inductive proximity sensors include:

- No moving parts/no mechanical wear.
- Not color dependent.
- Less surface dependent than other sensing technologies.

The disadvantages of inductive proximity sensors include:

- Only sense the presence of metal targets.
- Operating range is shorter than ranges available in others sensing technologies.
- Maybe affected by strong electromagnetic fields.

2. Capacitive Proximity Sensors

Capacitive proximity sensor is a noncontact technology suitable for detecting metals, non- metals such as paper, glass, liquids and cloth, Figure 1.3. However, it is best suited for nonmetallic targets because of its characteristics and cost relative to inductive proximity sensors. In most applications with metallic targets, inductive sensing is preferred because it is both a reliable and a more affordable technology.

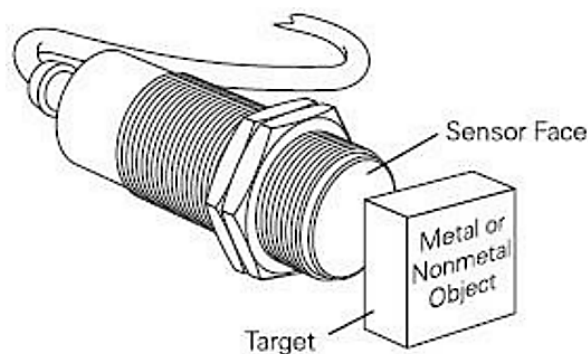


Figure 1.3: Capacitive Proximity Sensor

Capacitive proximity sensors consist of four main components: capacitive probe or plate, oscillator, signal level detector, output switching device, Figure 1.4. These sensors are similar in size, shape, and concept to inductive proximity sensors. However, capacitive proximity sensors react to alterations in an electrostatic field. The probe behind the sensor face is a capacitor plate. When power is applied to the sensor, an electrostatic field is generated that reacts to changes in capacitance caused by the presence of a target. When the target is outside the electrostatic field, the oscillator is inactive. As the target approaches, a capacitive coupling develops between the target and the capacitive probe. When the capacitance reaches a specified threshold, the oscillator is activated, triggering the output circuit to switch states between ON or OFF, Figure 1.4.

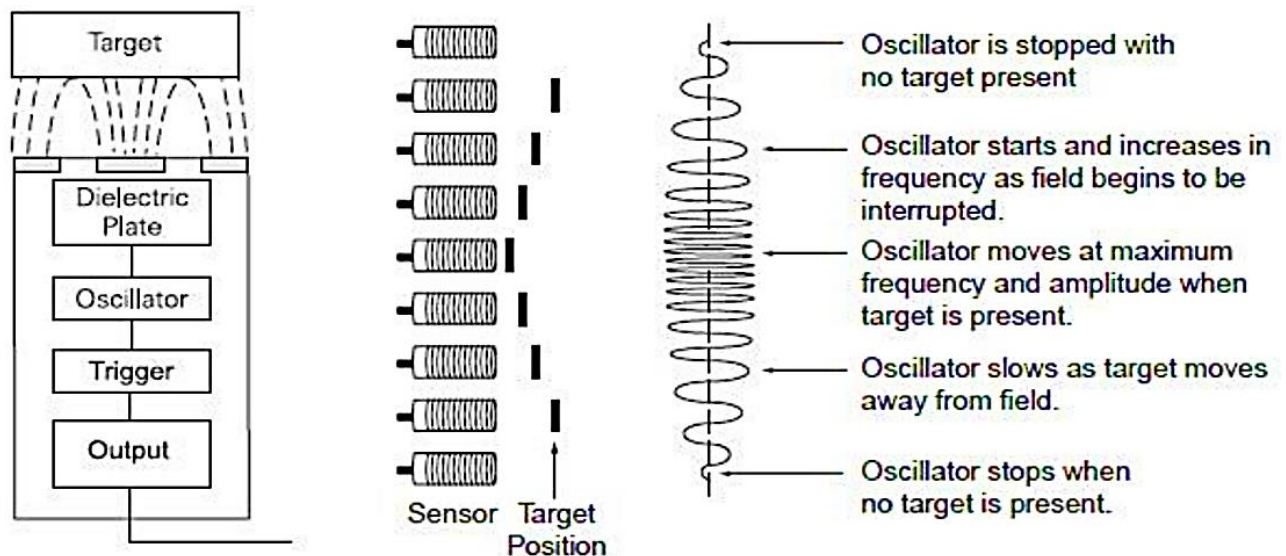


Figure 1.4: Capacitive Proximity Sensor Set-Up and Operation

An important point to be considered while using capacitive proximity sensor is that any material entering the sensor's electrostatic field can cause an output signal. This includes mist, dirt, dust, or other contaminants on the sensor face.

The advantages of capacitive proximity sensors include:

- Detects metal and nonmetal, liquids and solids. - Can “see through” certain materials (product boxes).
- Solid-state, long life.
- Many mounting configurations.

The disadvantages of capacitive proximity sensors include:

- Short (1 inch or less) sensing distance varies widely according to material being sensed.
- Very sensitive to environmental factors (humidity in coastal/water climates can affect sensing output).
- Not at all selective for its target, hence, control of what comes close to the sensor is essential.

One application for capacitive proximity sensors is level detection through a barrier. For example, water has a much higher dielectric than plastic. This gives the sensor the ability to “see through” the plastic and detect the level water, Figure 1.5.



Figure 1.5: Application for Capacitive Proximity Sensors

3. Magnetic Proximity Sensors

Magnetic proximity sensors are noncontact proximity devices utilize Hall Effect principles. Magnetic proximity sensors are characterized by the possibility of large switching distances and availability with small dimensions. They detect magnetic objects (usually permanent magnets), which are used to trigger the switching process. Magnetic proximity sensors are actuated by the presence of a permanent magnet. Their operating principle is based on the use of “reed contacts”, Figure 1.6, which are thin plates hermetically sealed in a glass bulb with inert gas. The presence of a magnetic field forces the thin plates to touch each other causing an electrical contact. The surface of plate has been treated with a special material particularly suitable for low current or high inductive circuits.

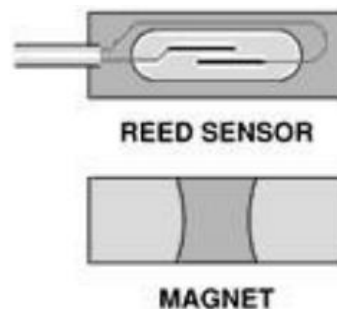


Figure 1.6: Hysteresis in Magnetic Proximity Sensor

The advantages of magnetic sensors compared to traditional mechanical switches are the following:

- Contacts are well protected against dust, oxidization and corrosion due to the hermetic glass bulb and inert gas; contacts are activated by means of a magnetic field rather than mechanical parts.
- Special surface treatment of contacts assures long contact life.
- Maintenance free. - Easy operation and small size

4. Optical Proximity Sensors

In its most basic form, a photoelectric sensor can be thought of as a switch where the mechanical actuator or lever arm function is replaced by a beam of light. By replacing the lever arm with a light beam the device can be used in applications requiring sensing distances from less than 2.54 cm to one hundred meters or more. All photoelectric sensors operate by sensing a change in the amount of light received by a photo detector. The change in light allows the sensor to detect the presence or absence of the object, its size, shape, reflectivity, opacity, translucence, or color. There are a vast number of photoelectric sensors from which to choose. Each offers a unique combination of sensing performance, output characteristics, and mounting options. A photoelectric sensor consists of five basic components: light source, light detector, lenses, logic circuit, and the output, Figure 1.7. A light source sends light toward the object. A light receiver, pointed toward the same object, detects the presence or absence of direct or reflected light originating from the source. Detection of the light generates an output signal (analog or digital).

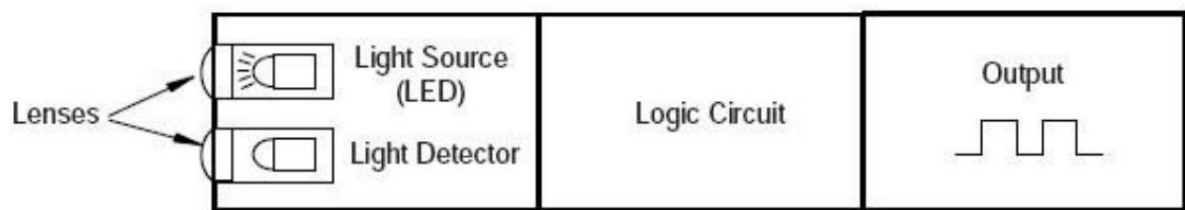


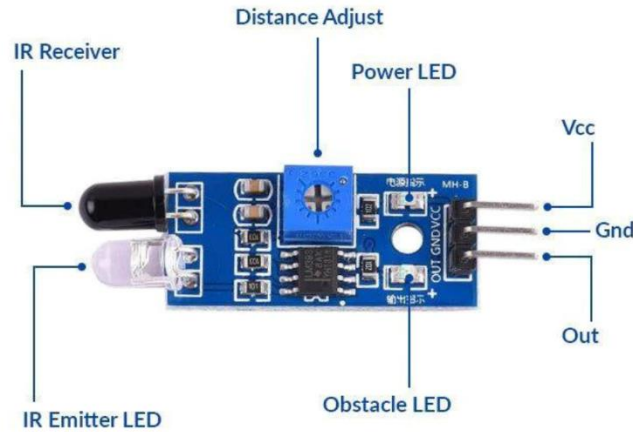
Figure 1.7: Optical Proximity Sensor

Photoelectric sensors can be housed in separate source and receiver packages or as a single unit. An important part of any sensor application involves selecting the best sensing mode for the application. There are three basic types of sensing modes in photoelectric sensors: Transmitted beam, Retroreflective, and Diffuse sensors.

Infrared sensor

An infrared sensor is an electronic device, that emits in order to sense some aspects of the surroundings. An IR sensor can measure the heat of an object as well as detects the motion. These types of sensors measure only infrared radiation, rather than emitting it that is called a passive IR sensor. Usually, in the infrared spectrum, all the objects radiate some form of thermal radiation.

These types of radiations are invisible to our eyes, which can be detected by an infrared sensor. The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode that is sensitive to IR light of the same wavelength as that emitted by the IR LED. When IR light falls on the photodiode, the resistances and the output voltages will change in proportion to the magnitude of the IR light received.



Working Principle:

The working principle of an infrared sensor is similar to the object detection sensor. This sensor includes an IR LED & an IR Photodiode, so by combining these two can be formed as a photo-coupler otherwise optocoupler. The physics laws used in this sensor are planks radiation, Stephan Boltzmann & weins displacement.

IR LED is one kind of transmitter that emits IR radiations. This LED looks similar to a standard LED and the radiation which is generated by this is not visible to the human eye. Infrared receivers mainly detect the radiation using an infrared transmitter. These infrared receivers are available in photodiodes form. IR Photodiodes are dissimilar as compared with usual photodiodes because they detect simply IR radiation. Different kinds of infrared receivers mainly exist depending on the voltage, wavelength, package, etc.

Once it is used as the combination of an IR transmitter & receiver, then the receiver's wavelength must equal the transmitter. Here, the transmitter is IR LED whereas the receiver is IR photodiode. The infrared photodiode is responsive to the infrared light that is generated through an infrared LED. The resistance of photo-diode & the change in output voltage is in proportion to the infrared light obtained. This is the IR sensor's fundamental working principle.

Once the infrared transmitter generates emission, then it arrives at the object & some of the emission will reflect back toward the infrared receiver. The sensor output can be decided by the IR receiver depending on the intensity of the response.

Types of Infrared Sensor

Infrared sensors are classified into two types like active IR sensor and passive IR sensor.

Active IR Sensor:

This active infrared sensor includes both the transmitter as well as the receiver. In most of the applications, the light-emitting diode is used as a source. LED is used as a non-imaging infrared sensor whereas the laser diode is used as an imaging infrared sensor. These sensors work through energy radiation, received & detected through radiation. Further, it can be processed by using the signal processor to fetch the necessary information. The best examples of this active infrared sensor are reflectance and break beam sensor.

- ✓ This type of sensor includes both the emitter & the receiver, which are also known as the transmitter & receiver. In the majority of cases, a laser diode or LED is used as a light source. LEDs are used for non-imaging infrared sensors, while laser diodes are used for imaging infrared sensors. This sensor works by emitting energy that is detected and received by the detector. To obtain the required data, it is further processed through a signal processor. Reflectance and break beam sensors are the best examples of active infrared detectors.



Passive IR Sensor:

The passive infrared sensor includes detectors only but they don't include a transmitter. These sensors use an object like a transmitter or IR source. This object emits energy and detects through infrared receivers. After that, a signal processor is used to understand the signal to obtain the required information. The best examples of this sensor are pyroelectric detector, bolometer, thermocouple-thermopile, etc. These sensors are classified into two types like thermal IR sensor and quantum IR sensor. The thermal IR sensor doesn't depend on wavelength. The energy source used by these sensors is heated. Thermal detectors are slow with their response and detection time. The quantum IR sensor depends on the wavelength and these sensors include high response and detection time. These sensors need regular cooling for specific measurements.

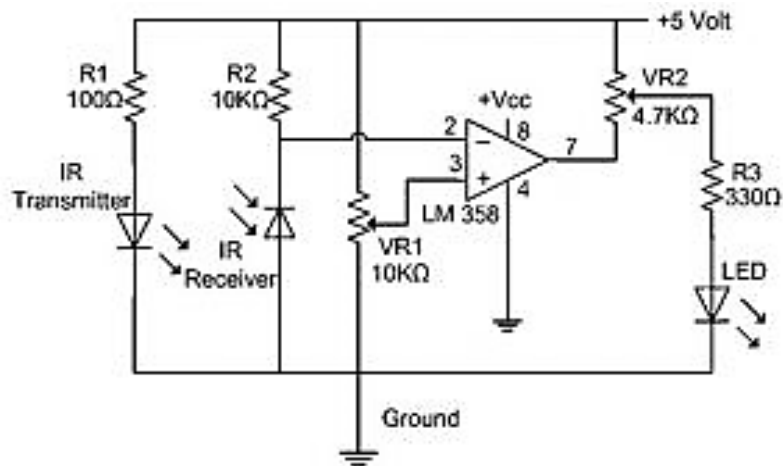
- ✓ The passive infrared sensor includes detectors only, but they don't include a transmitter. These sensors make use of a transmitter or an infrared source. This object emits energy, which infrared receivers detect. After that, a signal processor is used to decode the signal and extract the necessary data.



IR Sensor Circuit Diagram: PROJECT-1

An infrared sensor circuit is one of the basic and popular sensor modules in an electronic device. This sensor is analogous to human's visionary senses, which can be used to detect obstacles and it is one of the common applications in real-time. This circuit comprises the following components

- ✓ LM358 IC 2 IR transmitter and receiver pair
- ✓ Resistors of the range of kilo-ohms.
- ✓ Variable resistors.
- ✓ LED (Light Emitting Diode).



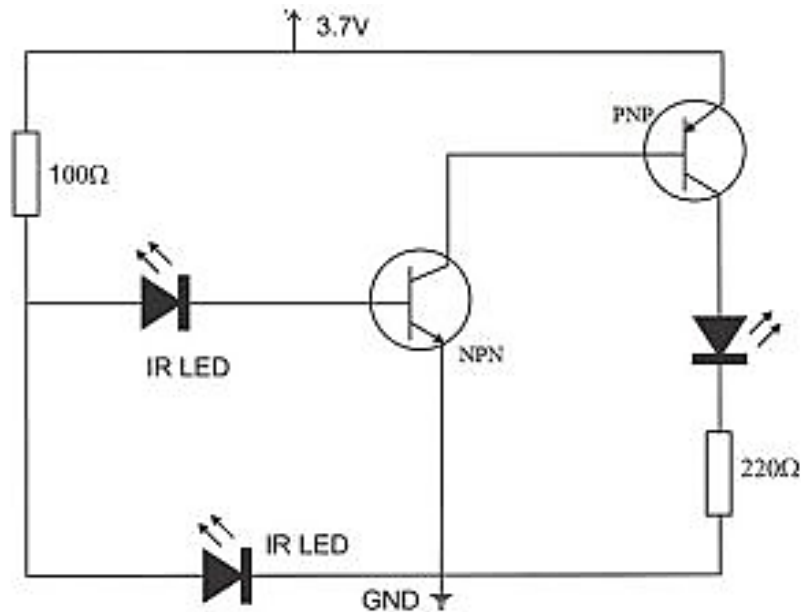
In this project, the transmitter section includes an IR sensor, which transmits continuous IR rays to be received by an IR receiver module. An IR output terminal of the receiver varies depending upon its receiving of IR rays. Since this variation cannot be analyzed as such, therefore this output can be fed to a comparator circuit. Here an operational amplifier (op-amp) of LM 339 is used as a comparator circuit.

When the IR receiver does not receive a signal, the potential at the inverting input goes higher than that non-inverting input of the comparator IC (LM339). Thus the output of the comparator goes low, but the LED does not glow. When the IR receiver module receives a signal to the potential at the inverting input goes low. Thus the output of the comparator (LM 339) goes high and the LED starts glowing.

Resistor R1 (100), R2 (10k), and R3 (330) are used to ensure that a minimum of 10 mA current passes through the IR LED Devices like Photodiode and normal LEDs respectively. Resistor VR2 (preset=5k) is used to adjust the output terminals. Resistor VR1 (preset=10k) is used to set the sensitivity of the circuit Diagram.

IR Sensor Circuit using Transistor: PROJECT-2

The circuit diagram of the IR sensor using transistors namely obstacle detection using two transistors is shown below. This circuit is mainly used for obstacle detection using an IR LED. So, this circuit can be built with two transistors like NPN and PNP. For NPN, BC547 transistor is used whereas, for PNP, BC557 transistor is used. The pinout of these transistors is the same.



In the above circuit, one infrared LED is always switched on whereas the other infrared LED is allied to the PNP transistor's base terminal because this IR LED acts as the detector. The required components of this IR sensor circuit include resistors 100 ohms & 200 ohms, BC547 & BC557 transistors, LED, IR LEDs-2. The step by step procedure of how to make the IR sensor circuit includes the following steps.

Circuit Working: Once the infrared LED is detected, then the reflected light from the thing will activate a small current that will supply throughout the IR LED detector. This will activate the NPN transistor & the PNP; therefore the LED will switch ON. This circuit is applicable for making different projects like automatic lamps to activate once a person approaches close to the light.

Advantages: The advantages of IR sensor include the following

- It uses less power
- The detection of motion is possible in the presence or absence of light approximately with equal reliability.
- They do not need contact with the object for detection
- There is no data leakage because of the ray direction
- These sensors are not affected by oxidation & corrosion
- Noise immunity is very strong

Disadvantages: The disadvantages of IR sensor include the following

- Line of sight is required
- Range is limited
- These can be affected by fog, rain, dust, etc
- Less data transmission rate

- IR-based Sensors are prone to interference from heat sources such as direct sunlight or high-powered lamps, which can cause readings to be inaccurate or unreliable.
- The IR sensor has a limited signal range and can fail to deliver an accurate reading over long distances or through walls or other obstacles that might block the signal route between the system's transmitter and receiver elements.
- High humidity levels in the surroundings where these sensors work may also cause problems with their operation, which might result in false-positive readings even when no actual motion occurs within the viewable distance limits of this sort of device.
- Reliability issues due to rapid ageing if subjected to constantly changing temperatures, affecting how well they track movement over time, could lead to less than satisfactory results during use in outdoor locations exposed directly to extreme temperature fluctuation conditions like climates being close to Sea/Ocean surroundings.

Application:

IR sensors have found their applications in most of today's equipment. Following is the list of sensors that are named after their usage. It helps you to understand what an IR sensor is used for.

1. Proximity Sensor: These are used in cellphones to measure the distance between objects. These are used in mobile phones to determine the distance between two objects. They use a principle called Reflective Indirect Incidence.

The receiver receives radiation broadcast by the transmitter after it has been reflected by an object. Distance is calculated using the intensity of the radiation received.

2. Item Counter: This uses the direct incidence approach to count the items. The transmitter and receiver maintain a consistent radiation level.

As soon as the object cuts the radiation, the item is detected and the count is increased. The same count is shown on a display system.

3. Burglar Alarm: This is one of the widely used and commonly used sensor applications. It is another example of the direct incidence method.

It works similarly to the item counter, where the transmitter and receiver are kept on both sides of a door frame. Constant radiation is maintained between the transmitter and receiver; whenever an object crosses the path alarm starts.

4. Radiation Thermometers: It is one of the key Infrared sensor applications. The working of a radiation thermometer depends on temperature and the type of object.

These have faster response and easy pattern measurements. They can do measurements without direct contact with an object.

5. Human Body Detection: This method is used in intrusion detection, auto light switches, etc. An intrusion alarm system senses the temperature of the human body.

If the temperature is more than the threshold value, it sets the alarms. It uses an electromagnetic system that is suitable for the human body to protect it from unwanted harmful radiation.

6. Gas Analyzers: Gas analyzers use infrared absorption properties to assess gas density. There are dispersive and non-dispersive gas analyzers available.

7. Other applications: These sensors are also used in infrared imaging, optical power meters, sorting machines, missile guidance, remote sensing, flame monitors, moisture analyzers, and night vision.



Radiation Thermometers

Night Vision Devices

Gas Analyzers

IR Imaging Devices

Pulse oximeter

A pulse oximeter is a non-invasive medical device used to measure the oxygen saturation (SpO_2) of arterial blood and pulse rate. It is widely used in hospitals, ICUs, operation theatres, and home monitoring. A pulse oximeter works by clipping onto a body part, usually a fingertip, and shining two wavelengths of light (red and infrared) through the tissue to measure blood oxygen levels (SpO_2) and pulse. Oxygenated hemoglobin (O_2Hb) and deoxygenated hemoglobin (HHb) absorb these different light wavelengths differently; the oximeter detects how much light passes through, calculates the ratio, and displays your SpO_2 percentage and heart rate.

Oxygen saturation (SpO_2) is the percentage of hemoglobin molecules in arterial blood that are saturated with oxygen.

Principle:

The pulse oximeter works on the principles of:

1. Beer–Lambert Law (light absorption)
2. Photoplethysmography (pulsatile blood flow detection)

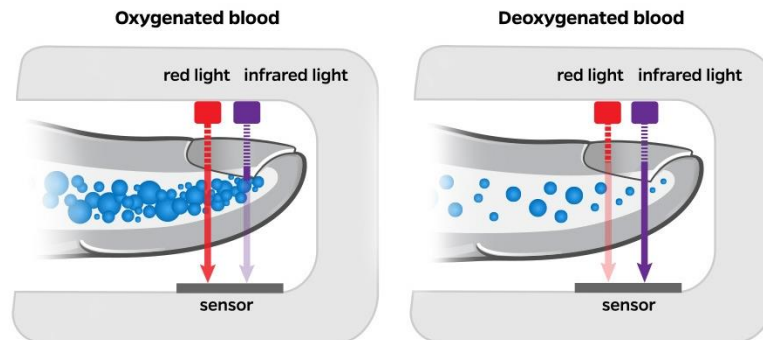
Oxygenated and deoxygenated hemoglobin absorb light differently at specific wavelengths.

Construction (Main Components):

- Light sources (LEDs):
 - Red light ($\approx 660 \text{ nm}$)
 - Infrared light ($\approx 940 \text{ nm}$)
- Photodetector

- Microprocessor
- Display unit
- Probe (finger/earlobe type)

Working Principle:



1. Emission of Light: The probe emits alternating red and infrared light through a vascular tissue such as a finger.
2. Absorption by Haemoglobin
 - Oxyhaemoglobin (HbO₂): Absorbs more infrared light and less red light
 - Deoxyhaemoglobin (Hb): Absorbs more red light and less infrared light
3. Detection of Transmitted Light: The photodetector measures the amount of red and infrared light that passes through the tissue.
4. Pulsatile Signal Isolation: The device separates the pulsatile component (arterial blood) from non-pulsatile components (skin, bone, venous blood).
5. Calculation of SpO₂: The ratio of absorbed red and infrared light is calculated and compared with stored calibration data to determine oxygen saturation.

$$\text{Ratio} = \frac{\text{Absorption of Red light}}{\text{Absorption of Infrared light}}$$

This ratio is compared with stored calibration data to compute oxygen saturation (SpO₂ %).

6. Display of Results: The pulse oximeter displays:
 - Oxygen saturation (SpO₂ %)
 - Pulse rate (beats per minute)

Normal Values:

- SpO₂: 95–100%
- Pulse rate: 60–100 beats/min (adult)

Advantages:

- Non-invasive and painless
- Continuous and real-time monitoring
- Easy to use and portable

Limitations:

- Inaccurate with poor peripheral perfusion
- Errors due to motion, nail polish, artificial nails
- Does not measure ventilation or carbon dioxide levels

Applications:

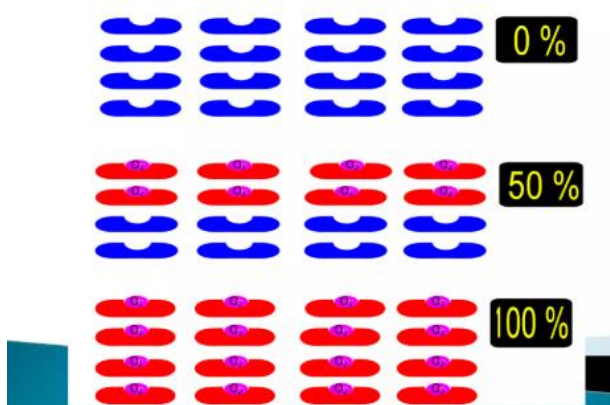
- Monitoring patients in ICUs and OT
- Emergency and anesthesia monitoring
- Home monitoring of respiratory diseases

Pulse Oximeter:

Pulse Oximeter used to measure oxygen saturation in the body i.e. how much of the hemoglobin in the blood is carrying the oxygen.



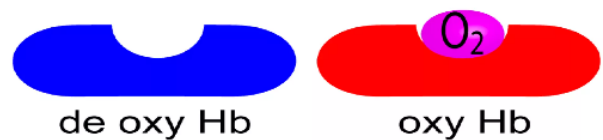
Ratio of saturation:



Oxygen Saturation:

Oxygen enters the lungs and then is passed on into blood. The blood carries the oxygen to the various organs in our body. The main way oxygen is carried in our blood is by means of hemoglobin.

Normal oxygen saturation values are 97% to 99% in a healthy individual on room air.



Working Principle:

- ❖ Pulse Oximetry consists of Red(R) and Infrared(IR) light emitting LEDs and a photo detector.
- ❖ Oxygenated and deoxygenated hemoglobin have different light absorption rate.
 - Oxygenated hemoglobin absorbs more infrared light
 - Deoxygenated hemoglobin absorbs more red light

Calculation:

$$\text{Oxygen saturation} = \frac{C(\text{HbO}_2)}{C(\text{HbO}_2) + C(\text{Hb})} \times 100 (\%)$$

$C(\text{Hb})$ = Concentration of deoxygenated hemoglobin

$C(\text{HbO}_2)$ = Concentration of oxygenated hemoglobin

Operation:

Finger is placed in between the light source and the light detector. Non absorbed light by finger reaches at detector.

Light is emitted from light sources which goes across the pulse oximeter probe and reaches the light detector.

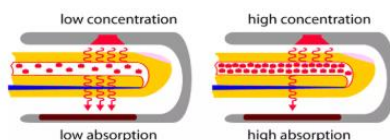


The amount of light absorbed depends on three physical properties:

1. concentration of the light absorbing substance.
2. length of the light path in the absorbing substance
3. oxyhemoglobin and deoxyhemoglobin absorbs red and infrared light differently

Property No.1

Amount of light absorbed is proportional to the concentration of the light absorbing substance.



Property No.2

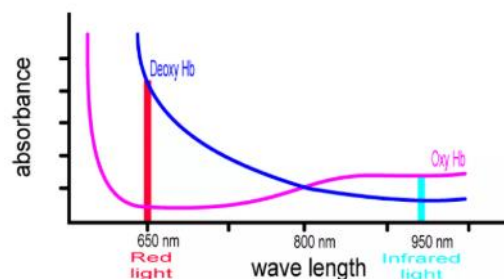
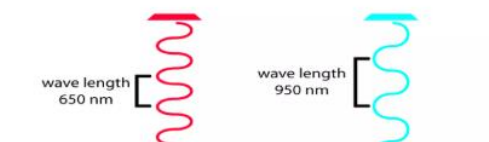
Amount of light absorbed is proportional to the length of the light path.

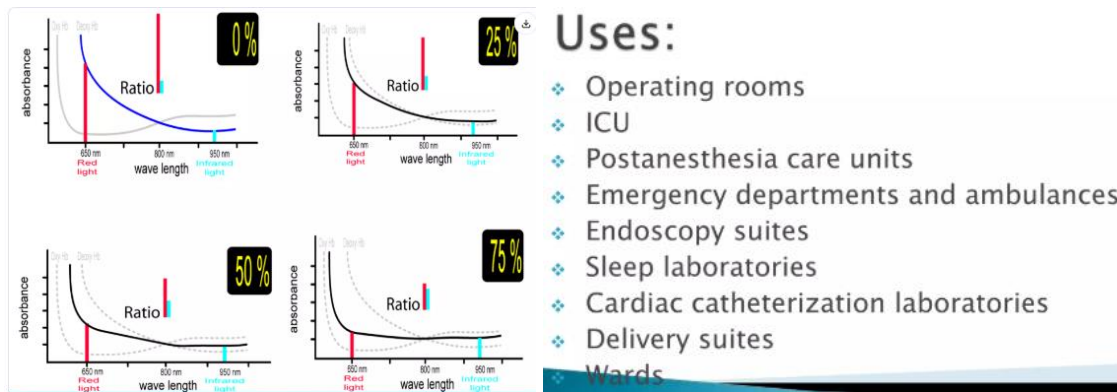


Using this ratio, the pulse oximeter can then work out the oxygen saturation.

Property No.3

oxyhemoglobin absorbs more infrared light than red light & deoxyhemoglobin absorbs more red light than infrared light.





Tachometer

Introduction: A tachometer is a device that measures the speed of rotation of an object and displays it in RPM or angular velocity units. A tachometer is an instrument used to measure the rotational speed of a shaft or disk, usually expressed in revolutions per minute (RPM). It is widely used in automobiles, machines, turbines, and laboratory equipment to monitor and control speed.

Principle of Operation: The working principle of a tachometer depends on its type, but generally it converts rotational motion into a measurable signal (mechanical, electrical, or optical) proportional to speed.

Types of Tachometers:

A. Mechanical Tachometer

- Uses gears and flexible shafts
- Speed is indicated by a needle on a dial
- Example: Older vehicles

B. Electrical (Electronic) Tachometer

- Works on electromagnetic or pulse-counting principles
- Converts speed into electrical signals
- More accurate than mechanical types

C. Optical (Photo) Tachometer

- Uses a light beam and reflective strip
- Counts interruptions of light to calculate RPM
- Non-contact type

D. Digital Tachometer

- Displays speed digitally

- Often uses sensors and microcontrollers
- High accuracy and easy reading

5. Construction

A typical tachometer consists of:

- Sensing element (gear, magnetic pickup, or optical sensor)
- Signal processing unit
- Indicating unit (dial or digital display)
- Power supply (for electronic types)

6. Working

- The rotating shaft generates a signal (mechanical motion, electrical pulse, or light interruption).
- This signal is processed and converted into speed data.
- The result is displayed as RPM on the indicator.

7. Applications

- Automobiles (engine speed measurement)
- Industrial machines
- Aircraft and ships
- Power plants and turbines
- Laboratory experiments

8. Advantages

- Accurate speed measurement
- Helps prevent overspeeding
- Improves machine efficiency and safety
- Available in contact and non-contact types

9. Limitations / Errors

- Mechanical wear (in mechanical tachometers)
- Signal noise in electronic tachometers
- Alignment and reflection errors in optical tachometers

10. Conclusion

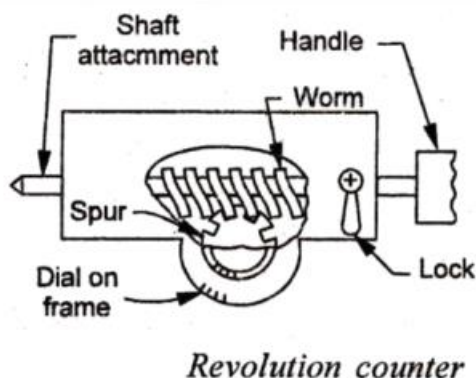
A tachometer is an essential instrument for monitoring rotational speed in mechanical and electrical systems. With advancements in technology, modern digital and optical tachometers provide high accuracy, reliability, and ease of use.

Tachometers may be broadly classified into two categories:

1. Mechanical tachometers.
2. Electrical tachometers.

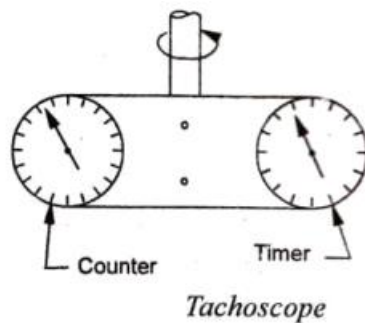
Mechanical tachometers: These tachometers employ only mechanical parts and mechanical movements for the measurement of speed.

Revolution counter and timer: The revolution counter, sometimes called a speed counter, consists of a worm gear which is also the shaft attachment and is driven by the speed source. The worm drives the spur gear which in turn actuates the pointer on a calibrated dial. The pointer indicates the number of revolutions turned by the input shaft in a certain length of time. The unit requires a separate timer to measure the time interval. The revolution counter, thus, gives an average rotational speed rather than an instantaneous rotational speed. Such speed counters are limited to low speed engines which permit reading the counter at definite time intervals. A properly designed and manufactured revolution counter would give a satisfactory speed measure upto 2000-3000 rpm.

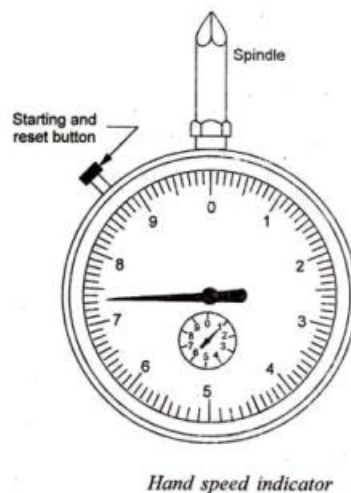


2. Tachoscope: The difficulty of starting a counter and a watch at exactly the same time led to the development of tachoscope, which consists of a revolution counter incorporating a built-in timing device. The two components are integrally mounted, and start simultaneously when the contact point is pressed against the rotating shaft. The instrument runs until the contact point is disengaged from

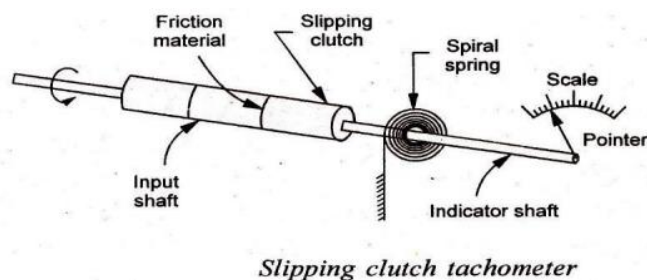
the shaft. The rotational speed is computed from the readings of the counter and timer. Tachoscopes have been used to measure speeds upto 5000 rpm.



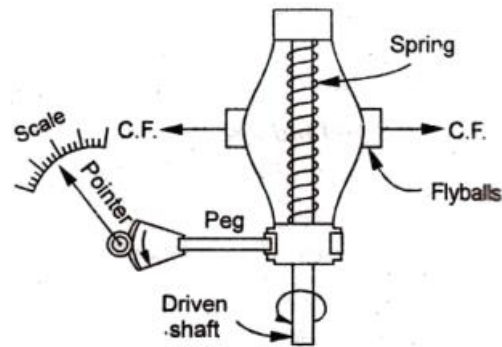
3. Hand speed indicator: The indicator has an integral stop watch and counter with automatic disconnect. The spindle operates when brought in contact with the shaft, but the counter does not function until the start and wind button is pressed to start the watch and engage the automatic clutch. Depressing of the starting button also serves to wind the starting watch. After a fixed time interval (usually 3 or 6 seconds), the revolution counter automatically gets disengaged. The instrument indicates the average speed over the short interval, and the dial is designed to 88 indicate the rotational speed directly in rpm. These speed measuring units have an accuracy of about 1% of the full scale and have been used for speeds within the range 20,000 to 30,000 rpm.



4. Slipping clutch tachometer: The rotating shaft drives an indicating shaft through a slipping clutch. A pointer attached to the indicator shaft moves over a calibrated scale against the torque of a spring. The pointer position gives a measure of the shaft speed.

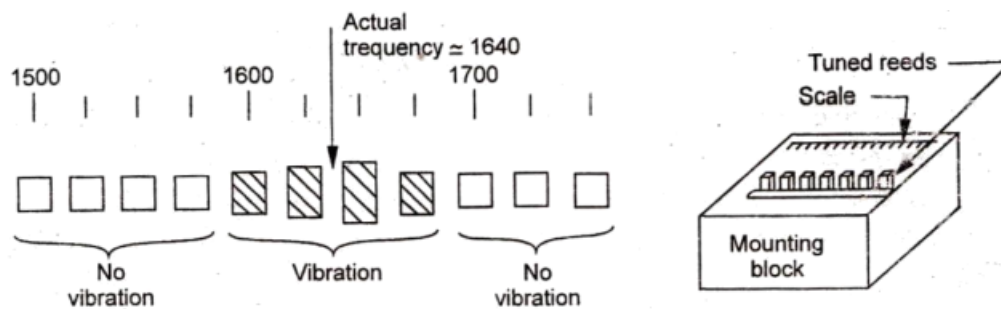


5. Centrifugal force tachometers: The device operates on the principle that centrifugal force is proportional to the speed of rotation. Two flyballs (small weights) are arranged about a central spindle. Centrifugal force developed by these rotating balls works to compress the spring as a function of rotational speed. A grooved collar or sleeve attached to the free end of the spring then slides on the spindle and its position can be calibrated in terms of the shaft speed. Through a series of linkages, the motion of the sleeve is usually amplified and communicated to the pointer of the instrument to indicate speed. Certain attachments can be mounted onto the spindle to use these tachometers for the measurement of linear speed.



Centrifugal tachometer

6. Vibrating reed tachometer: Tachometers of the vibrating reed type utilize the fact that speed and vibration in a body are interrelated. The instrument consists of a set of vertical reeds, each having its own natural frequency of vibration. The reeds are lined up in order of their natural frequency and are fastened to a base plate at one end, with the other end free to vibrate. When the tachometer base plate is placed in mechanical contact with the frame of a rotating machine, a reed tuned to resonance with the machine vibrations responds most frequently. The indicated reed vibration frequency can be calibrated to indicate the speed of the rotating machine.



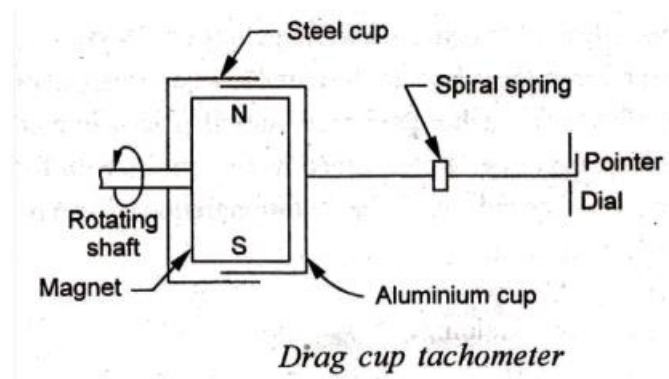
Vibrating reed tachometer

Electrical tachometers: An electrical tachometer depends for its indications upon an electrical signal generated in proportion to the rotational speed of the shaft. Depending on the type of the transducer, electrical tachometers have been constructed in a variety of different designs.

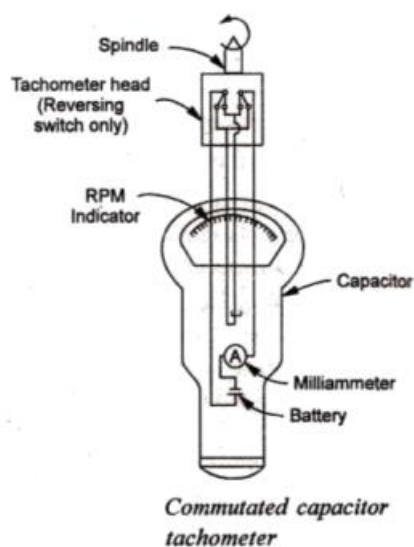
1. Drag cup tachometer:

In an eddy current or drag type tachometer, the test shaft rotates a permanent magnet and this induces eddy currents in a drag cup or disc held close to the magnet. The eddy currents produce a torque

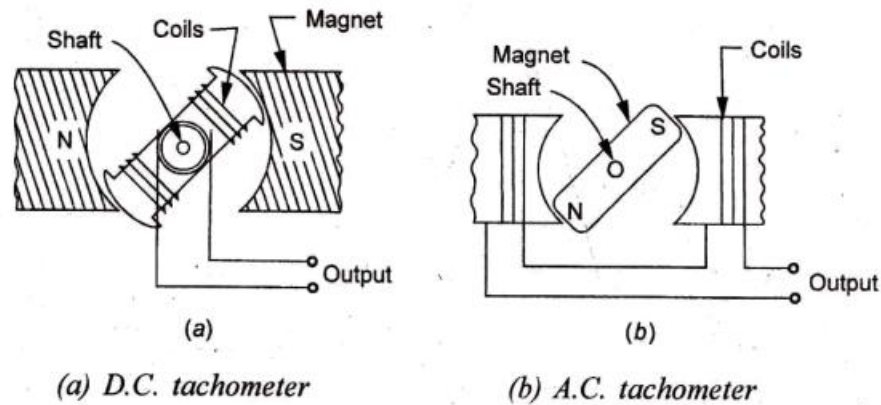
which rotates the cup against the torque of a spiral spring. The disc turns in the direction of the rotating magnetic field until the torque developed equals that of the spring. A pointer attached to the cup indicates the rotational speed on a calibrated scale. The automobile speedometers operate on this principle and measure the angular speed of the wheels. The rotational measurement is subsequently converted into linear measurement by assuming some average diameter of the wheel, and the scale is directly calibrated in linear speed units. Eddy current tachometers are used for measuring rotational speeds upto 12,000 rpm with an accuracy of $\pm 3\%$.



2. Commutated capacitor tachometer: The operation of this tachometer is based on alternately charging and discharging a capacitor. These operations are controlled by the speed of the machine under test. The instrument essentially consists of: h) Tachometer head containing a reversing switch, operated by a spindle which reverses twice with each revolution. 10. Indicating unit containing a voltage source, a capacitor, milliammeter and a calibrating circuit. When the switch is closed in one direction, the capacitor gets charged from d-c supply and the current starts flowing through the ammeter. When the spindle operates the reversing switch to close it in opposite direction, capacitor discharges through the ammeter with the current flow direction remaining the same. The instrument is so designed that the indicator responds to the average current. Thus, the indications are proportional to the rate of reversal of contacts, which in turn are proportional to speed of the shaft. The meter scale is graduated to read in rpm rather than in milliamperes. The tachometer is used within the range 200 - 10000 rpm.



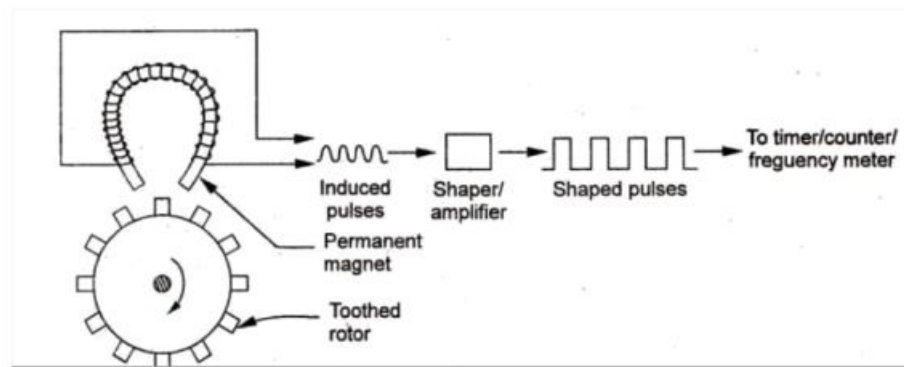
10) Tachogenerators: These tachometers employ small magnet type d.c or a.c generators which translate the rotational speeds into d.c. or a.c voltage signal. The operating principle of such tachometers is illustrated in Fig. Relative perpendicular motion between a magnetic field and conductor results in voltage generation in the conductor.



10) **D. C. tachometer generator:** This is an accurately made dc. generator with a permanent magnet of horse-shoe type. With rotation of the shaft, a pulsating dc. Voltage proportional to the shaft speed is produced, and measured with the help of a moving coil voltmeter having uniform scale and calibrated directly in terms of speed. The tachometer is sensitive to the direction of rotation and thus can be used to indicate this direction by the use of an indicator with its zero point at mid-scale. For greater accuracy, air gap of the magnetic paths must be maintained as uniform as possible. Further, the instrument requires some form of commutation which presents the problem of brush maintenance.

11) **A.C. tachometer generator:** The unit embodies a stator surrounding a rotating permanent magnet. The stator consists of a multiple pole piece (generally four), and the permanent magnet is installed in the shaft whose speed is being measured. When the magnet rotates, an a.c. voltage is induced in the stator coil. The output voltage is rectified and measured with a permanent magnet moving coil instrument. The instrument can also be used to measure a difference in speed of two sources by differentially connecting the stator coils. Tachogenerators have been successfully employed for continuous measurement of speeds upto 500 rpm with an accuracy of $\pm 1\%$.

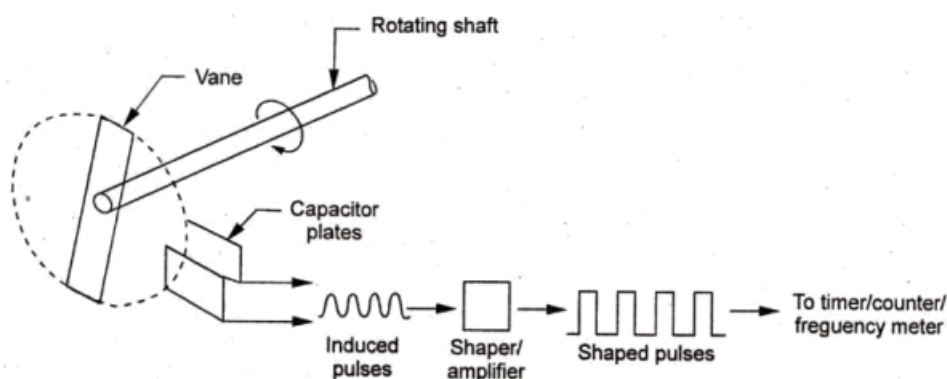
4. Contactless electrical tachometers: Tachometers of this type produce pulse from a rotating shaft without any physical contact between the speed transducer and the shaft. This aspect has the distinct advantage in that no load is applied to the machine. Inductive pick-up tachometer: The unit consists of a small permanent magnet with a coil round it. This magnetic pick up is placed near a metallic toothed rotor whose speed is to be measured. As the shaft rotates, the teeth pass in front of the pick-up and produce a change in the reluctance of the magnetic circuit. The field expands or collapses and a voltage is induced in the coil. The frequency of the pulses depends upon the number of teeth on the wheel and its speed of rotation. Since the number of teeth is known, the speed of rotation can be determined by measuring the pulse frequency. To accomplish this task, pulse is amplified and squared, and fed into a counter of frequency measuring unit. If the rotor has 60 teeth, and if the counter counts the pulses in one second, then the counter will directly display the speed in revolutions per minute.



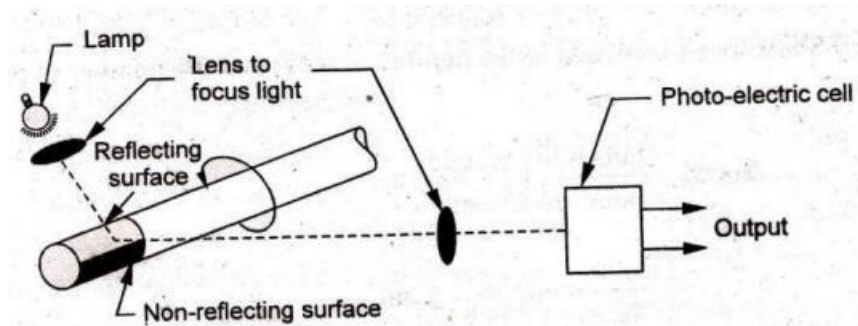
$$\text{Speed } N = \frac{\text{pulses per second}}{\text{number of teeth}}$$

$$N = \frac{P}{T} \text{ rps} = \frac{P}{T} \times 60 \text{ rpm}$$

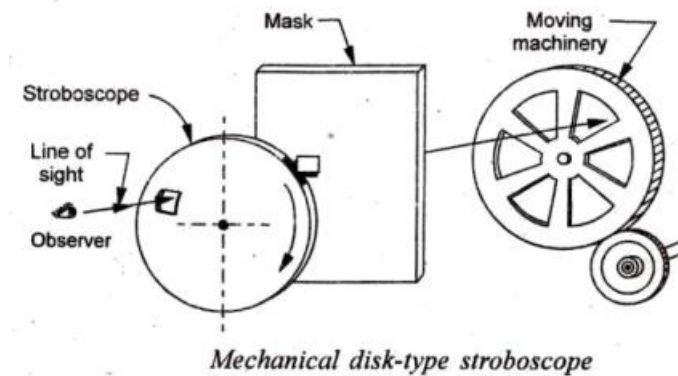
(ii) Capacitive type pick-up tachometer: The device consists of a vane attached to one end of the rotating machine shaft. When the shaft rotates between the fixed capacitive plates, there occurs a change in the capacitance. The capacitor forms a part of an oscillator tank so that number of frequency changes per unit of time is a measure of the shaft speed. The pulses thus produced are amplified, and squared, and may then be fed to frequency measuring unit or to a digital counter so as to provide a digital analog of the shaft rotation.



(iii) Photo-electric tachometer: These pick-ups utilize a rotating shaft to intercept a beam of light falling on a photo-electric or photo conductive cell. The shaft has an intermittent reflecting (white) and non-reflecting (black) surface. When a beam of light hits the reflecting surface on the rotating shaft, light pulses are obtained and the reflected light is focused onto the photo-electric cell. The frequency of light pulses is proportional to the shaft speed, and so will be the frequency of electrical output pulses from the photo-electric cell.



(iv)Stroboscope: The stroboscope utilises the phenomenon of vision when an object is viewed intermittently. The human sense of vision is so slow to react to light stimuli that it is unable to separate two different light impulses reaching the eye within a very short Period of time (less than 0.1second). A succession of impulses following one another at brief intervals are observed by the eye as a continuous unbroken sequence. A mechanical disk type stroboscope consists essentially of a whirling disk attached to motor whose speed can be varied and measured. A reference mark on the rotating shaft on the shaft appears to be stationary.



For this condition, the shaft speed equals that of rotating disk, or some even multiple of this speed and is given by:

$$\text{Shaft speed} = \frac{(\text{disk speed}) \times (\text{number of openings in the disk})}{\text{number of images}}$$